

Deep Learning for OCT-Based Classification of Visual Acuity in Patients Affected by Wet AMD

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خُدی کو کر بلند اتنا کہ ہر تقدیر سے پہلے
خُدا بندے سے خود پوچھے، بتا، تیری رضا کیا ہے
علامہ اقبال

A timeless reminder by Allama Iqbal: *“Elevate your being so high that before fate is written, God Himself asks — what is your wish?”*

Una riflessione poetica che invita ad ambizione e crescita: essere così completi e consapevoli che persino il destino si fermi ad ascoltare la nostra volontà.

Abstract

Age-related macular degeneration (AMD) is a leading cause of vision loss, with its wet (neovascular) form requiring frequent monitoring and individualized treatment decisions. Optical Coherence Tomography (OCT) is essential for visualizing retinal structure and guiding therapy, but interpreting subtle, longitudinal changes remains challenging. Using longitudinal OCT data from 271 patients (321 eyes over 8–14 years), this study develops deep learning approaches to classify visual acuity in wet age-related macular degeneration (AMD) patients using optical coherence tomography (OCT) scans through two complementary tasks. For static classification (predicting "good" vs "poor" acuity from individual annual scans), we fine-tuned three ImageNet-pretrained CNNs (ResNet18, ResNet34, DenseNet121) adapted for single-channel OCT input. In image-wise cross-validation, DenseNet121 achieved 83.7% accuracy, 82.9% F1 score, and 92.3% AUC, outperforming ResNet variants (81.7%/79.5% accuracy). However, performance dropped substantially in patient-wise evaluation (DenseNet121: 69.3% accuracy, 75.9% F1 score, 76.6% AUC), revealing the challenge of generalizing to unseen patients. For temporal classification (predicting year-over-year improvement), we developed a hybrid ResNet18-LSTM model processing sequential OCT features, comparing full-history and last-year-only strategies. Across all follow-up years, both approaches achieved modest average performance (typically 65–72% accuracy and AUC values around 55–60%), with no consistent benefit from using longer history. Grad-CAM visualizations confirmed that the static model focused on clinically relevant retinal features. This work demonstrates both the promise and limitations of deep learning for AMD acuity prediction, highlighting the importance of rigorous patient-level validation and offering insights for potential clinical integration.

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Chapter 1

Introduction

Age-related macular degeneration (AMD) is one of the leading causes of irreversible vision loss in older adults, especially in developed countries [1]. This chronic retinal disease comes in two forms: a **dry** (non-exudative) form and a more severe **wet** (exudative) form. The dry form accounts for about 90% of cases and typically causes gradual, progressive central vision loss. In contrast, the wet form (also called neovascular AMD, or wAMD) is characterized by the growth of abnormal blood vessels beneath the retina, which leak fluid or bleed, leading to scarring and a rapid deterioration of central vision [1]. Despite advances in treatments (notably anti-VEGF drug injections that can slow disease progression), wAMD remains a major cause of severe vision impairment. Its impact is growing with the aging population – for example, the global prevalence of AMD is projected to reach 288 million by 2040 [2] – underscoring the urgent need for effective management strategies.

Optical Coherence Tomography (OCT) has become an indispensable imaging modality in ophthalmology to diagnose and monitor retinal diseases such as AMD. OCT provides high-resolution cross-sectional images of the retina, allowing doctors to visualize retinal layers and detect pathological changes such as fluid accumulation. It is considered a fundamental tool in the diagnosis and management of wAMD, allowing the identification of choroidal neovascularization and treatment-related changes [3]. In routine wAMD care, patients undergo OCT scans at each clinic visit to guide treatment decisions (for example, determining whether an anti-VEGF injection is needed). However, while OCT produces a wealth of quantitative biomarkers (e.g., retinal thickness, lesion volume), interpreting disease progression from multiple OCT scans over time can be challenging. Clinicians must manually compare scans from different visits, a time-consuming and subjective process in which subtle predictive patterns may be overlooked [2]. This limitation presents an opportunity for artificial intelligence: if automated deep learning methods can discern meaningful trends in OCT data over time, they could assist ophthalmologists by forecasting disease

trajectory and visual outcomes, enabling earlier or more personalized interventions.

In recent years, AI and deep learning have started to revolutionize medical diagnostics, including applications in ophthalmology. Deep learning, particularly convolutional neural networks (CNNs), has achieved expert-level performance in detecting diseases from medical images. In ophthalmic diagnostics, deep learning systems have demonstrated high accuracy in identifying conditions like diabetic retinopathy and AMD from retinal photographs and OCT scans [4]. For example, a deep learning model developed by researchers at Google DeepMind was able to analyze 3D OCT scans and recommend referral decisions for over 50 eye diseases with about 94% accuracy – on par with expert clinicians [5]. Similarly, other studies have shown near-perfect accuracy in classifying common retinal pathologies from OCT images (distinguishing, for instance, between normal eyes, AMD, and diabetic macular edema) using very large datasets [6]. This success underscores the growing role of AI in ophthalmology: not only can it automate image classification with remarkable precision, but it also holds potential for more advanced tasks such as prognostication and outcome prediction [4].

While prior studies have demonstrated the promise of AI in ophthalmology, their problem formulations differ in clinically meaningful ways from ours. For instance, [4] provide a broad review highlighting AI’s utility in retinal disease progression analysis, supporting the relevance of our overall goal, but without offering a concrete implementation for OCT-based temporal prediction of visual acuity. [5], although impressive for classifying over 50 eye conditions and distinguishing wet vs. dry AMD using 3D OCT volumes, primarily addresses cross-sectional diagnosis at a single timepoint. Their use of full volumetric (3D) scans and advanced 3D CNNs differs from our focus on sequential 2D B-scans over time, reflecting practical differences in data availability and clinical workflows. Similarly, [6] tackle a temporal prediction task in wet AMD, but label the outcome as the need for an anti-VEGF injection—a direct reflection of treatment decisions, not functional vision improvement. While predicting injection need is simpler and operationally defined from records, we chose to predict BCVA change, aiming for a patient-centered outcome that is ultimately more meaningful to patients and clinicians. This choice introduces greater label noise and prediction difficulty but aligns with the clinical question of most interest: who will truly benefit functionally from ongoing treatment.

1.1 Motivation

While most existing ophthalmic AI tools focus on diagnosing disease or detecting features in a single scan, there is a crucial need to extend these capabilities to predict patient outcomes over time - a challenge we also address in this thesis by moving from static to temporal classification. In wAMD management, doctors face a difficult question: *what will a patient’s vision be in the future, and how*

will their disease respond to treatment?

Patients with wAMD require long-term therapy and frequent monitoring, yet outcomes can vary widely. Some patients maintain relatively good vision with prompt treatment, while others experience continuous decline despite therapy. Standard treatment regimens (e.g., fixed-interval injections or “treat-and-extend” protocols) are not one-size-fits-all – some individuals are over-treated and others under-treated, due to heterogeneous responses to therapy [2]. If we can accurately classify or predict which patients are likely to have a poor vision outcome versus a favorable outcome, clinicians could adjust treatment plans accordingly (for example, intervening more aggressively for a patient predicted to worsen, or spacing out visits for one predicted to remain stable). **Deep learning** is an attractive approach for this problem: by training on large numbers of OCT images with known patient outcomes, a neural network may discover subtle biomarkers or temporal patterns indicative of future vision loss or stability. Indeed, preliminary research has hinted at this possibility – for instance, deep learning has been applied to predict anti-VEGF treatment response in retinal disease from baseline OCT scans [6] and to forecast conversion from early/intermediate AMD to advanced wet AMD using single-timepoint imaging [6]. However, these early studies often rely on a single initial scan (static prediction) and do not fully exploit the rich longitudinal data available in a patient’s follow-up visits.

Another important consideration in clinical AI applications is explainability. Clinicians need to trust and understand the predictions of AI systems before integrating them into practice. Simply providing a classification is often insufficient; understanding why a model predicts “poor” or “good” outcome can help clinicians validate its reasoning and potentially discover new imaging biomarkers. To support this goal, methods such as Grad-CAM visualization can highlight which regions of an OCT scan most strongly influence the model’s prediction, offering insights into the features the network considers important. Including explainability mechanisms in AI models is thus an essential step toward building clinically usable, trustworthy tools.

1.2 Objectives

Given this background, the goal of this Master’s thesis is to investigate deep learning methods for **OCT-based classification of vision outcomes in wAMD**, leveraging both single-timepoint and sequential imaging data. In particular, we address three main objectives:

1. **Static classification** – Develop a deep learning model that takes OCT images of a year and classifies the vision outcome, which can be ‘Good’ or ‘Poor’.
2. **Explainability and interpretation (for static classification)** – Implement and evaluate explainability techniques, such as Grad-CAM visu-

alizations, to highlight the retinal features influencing the model’s predictions in the static classification setting.

3. **Temporal classification** – Develop a deep learning approach that analyzes a sequence of OCT images from the same eye over multiple visits (a time series spanning months or years) to predict or classify the patient’s visual outcome. This sequence-based prediction leverages the temporal evolution of the disease, potentially capturing trends (improvement or deterioration) that a single snapshot cannot.

These tasks are complementary. The static classification represents the scenario of making an early prognosis from an initial exam, which is valuable if we have only one timepoint. To address the need for trust and insight into the model’s decisions, we also incorporate explainability techniques (Grad-CAM) to highlight OCT features influencing the predictions. This helps bridge the gap between deep learning results and clinical understanding. The temporal classification represents the scenario of continuously learning from a patient’s history to refine predictions. By comparing these approaches, we can assess how much benefit is gained by incorporating longitudinal information. Throughout the thesis, we approach this problem from a computer engineering perspective, focusing on the design, training, and evaluation of neural network models – while maintaining an accessible description of medical concepts. We avoid excessive medical jargon and detail, aiming instead to bridge the gap between deep learning techniques and their medical application in a way that is understandable to both technical and clinical readers.

1.3 Thesis Outline

This thesis is organized as follows:

- **Chapter 1: Introduction** – Provides background on AMD and OCT imaging, reviews related work in AI-based ophthalmic diagnostics, motivates the need for predictive modeling of visual outcomes, and states the objectives of this research.
- **Chapter 2: Related Work** – Reviews existing methods and studies on automated OCT interpretation and visual acuity prediction, highlighting similarities and differences with our approach.
- **Chapter 3: Methodology** – Describes the data preprocessing, labeling strategies, model architectures (CNNs and LSTM), and training procedures used for both static and temporal classification tasks.
- **Chapter 4: Experiments and Results** – Presents the experimental setup, evaluation metrics, and results for static and temporal classification, including cross-validation strategies, performance comparisons, and Grad-CAM visualizations for explainability.

- **Chapter 5: Conclusion** – Summarizes the main findings, discusses limitations and challenges, and suggests directions for future work in deploying deep learning for OCT-based outcome prediction in AMD.

Chapter 2

Literature Review

In this chapter, a comprehensive overview of the current state of the art relevant to this thesis is presented, starting with an outline of the field of ophthalmology, with a focus on Age-related Macular Degeneration (AMD), particularly its neovascular form, known as wet AMD. Then the growing role of Artificial Intelligence (AI) and Deep Learning (DL) in medicine, highlighting their applications in clinical data analysis and medical imaging is explored. A special emphasis is placed on the use of Optical Coherence Tomography (OCT) in ophthalmology, and how deep learning has been successfully applied to analyze OCT scans for disease detection and outcome prediction. In addition, widely used deep learning architectures such as ResNet and DenseNet and their hybrid variants with LSTM networks are discussed, along with the importance of explainability in medical AI through techniques like Grad-CAM.

2.1 Ophthalmology and Age-Related Macular Degeneration (AMD)

There are many advancements going on in the field of medicine, and similarly, Ophthalmology—the branch of medicine that deals with eye and vision care—has also gone through advancements due to the integration of imaging technologies. One of those technologies is Optical Coherence Tomography (OCT). Ophthalmology plays an important role in diagnosing and managing diseases related to the eye and vision, and one of them is Age-related Macular Degeneration (AMD). AMD primarily affects the macula and is a leading cause of irreversible vision loss.

When AMD advances, it can be classified into two main types: dry (non-neovascular) and wet (neovascular). The dry form is more common but progresses slowly, whereas wet AMD, though less prevalent (10–15% of AMD cases), is responsible for the majority of severe vision impairments. Global prevalence

estimates suggest that 196 million people were affected by AMD in 2020, with projections rising to 288 million by 2040 [7].

Nowadays, Optical Coherence Tomography (OCT) is used to visualize biomarkers for wet AMD. These biomarkers help in identifying and monitoring the disease. Some key biomarkers include fluid localization (intraretinal fluid, sub-retinal fluid), structural features like hyperreflective foci, and changes in the ellipsoid zone. Best Corrected Visual Acuity (BCVA) is a key measure of visual function in individuals with AMD, and changes in BCVA can reflect the severity, progression, and response to treatment for this condition.

Patients affected by AMD are usually treated with anti-vascular endothelial growth factor (VEGF) treatments. Ranibizumab injections are commonly used to treat wet AMD. Late-stage wet AMD accounts for the majority of blindness cases and requires frequent anti-VEGF therapy, which poses economic and compliance challenges [8, 9].

The progressive nature of AMD, specially in its neovascular form, makes it important to diagnose early and monitor it continuously. This allows timely intervention for treatment, help in preserving visual function and improving the long term clinical outcomes. That’s why the OCT imaging and assessments, including BCVA are remain the best tools to track disease progression and guidance of personalized treatments.

2.2 Artificial Intelligence and Deep Learning in Medicine

In this era of technological advancement, where every field is influenced by Artificial Intelligence (AI), medicine is among those fields. AI is the ability of computer systems to perform those tasks that require human intelligence. The tasks performed by computer systems using the ability of AI includes learning, problem-solving, decision-making, and understanding language. In the AI, there is Machine Learning (ML) that focuses on algorithms and models that allows machines to learn from data. Then comes Deep Learning (DL) that is a subset of ML and it leverages neural networks with multiple layers to learn complex data representations.

AI and deep learning have permeated other clinical data domains. Electronic health records (EHRs) – containing structured data (demographics, vitals, lab results) and unstructured text (clinical notes) – are a rich source for predictive modeling. Deep neural networks have been trained on sequential EHR data to forecast outcomes such as disease onset, hospital readmission, or mortality [10]. Recurrent architectures (e.g. LSTMs and GRUs) were among the first to capture temporal patterns in longitudinal patient records, outperforming earlier statistical models on tasks like onset prediction [11]. Notably, an attention-based recurrent model (RETAIN) by Choi et al. learned to high-

light which past visits most influenced its predictions, achieving state-of-the-art results in heart failure risk stratification while improving interpretability [11]. In parallel, natural language processing of clinical narratives has advanced using transformer models. For example, researchers adapted BERT to healthcare (“ClinicalBERT”) to encode free-text notes, enabling better prediction of 30-day readmission from discharge summaries [11]. More recently, transformer-based architectures have broadly surpassed RNNs for EHR analytics: the BEHRT model (a BERT-inspired encoder of patient histories) was pre-trained on millions of primary care records and demonstrated an 8% absolute improvement in predicting future diagnoses across 300+ diseases compared to previous deep learning approaches [11]. These data-driven models can uncover subtle latent patterns in patient data, allowing clinicians to identify high-risk individuals and anticipate clinical events with greater accuracy [10].

Deep learning is also propelling breakthroughs in genomics and drug discovery. In genomics, advanced neural networks (including CNNs and attention-based models) can decipher high-dimensional “-omics” data, far exceeding manual analyses. Deep learning models now excel at variant calling and pathogenicity prediction from DNA sequencing data, splice-site and gene expression analysis, and even epitope–MHC binding predictions for immunogenomics [10]. These capabilities drive precision medicine by pinpointing disease-associated genetic factors and potential therapeutic targets. Likewise, AI has been embraced in pharmaceutical research: deep generative models and predictive frameworks are used to screen vast chemical libraries and design new compounds *in silico*. Such models significantly accelerate drug development by optimizing lead molecules and forecasting drug-target interactions or toxicity, improving efficiency and success rates [12]. Early examples show that AI-driven pipelines can shorten preclinical discovery from years to months while maintaining or increasing accuracy [13].

Researchers are integrating multiple data modalities with deep learning to inform clinical decisions, while grappling with unique challenges. Multimodal deep learning models combine data streams – for example, aggregating EHR tabular data, clinical text, lab results, and even medical images or genomics – to provide a more holistic patient representation [14]. Such systems can, for instance, fuse a patient’s imaging findings with their genetic markers and longitudinal clinical data to improve diagnostic or prognostic accuracy. Early multimodal studies report improved performance in tasks like outcome prediction by leveraging the complementary information in heterogeneous data [14]. These advances feed into intelligent clinical decision support systems that assist providers in risk assessment, diagnosis, and treatment selection [10]. Aided by deep learning, decision support tools can flag high-risk patients, forecast disease trajectories, and suggest personalized interventions based on a patient’s combined clinical profile [10]. However, deploying deep learning on broad healthcare data comes with challenges. Clinical data are often limited, noisy, or biased, which can hinder model training and generalization [10, 14]. Models also tend to operate as “black boxes,” and improving their interpretability is paramount in medicine

so that clinicians can trust and understand the recommendations [14]. Issues of data privacy, security, and integration further complicate development in this domain [10, 14]. Ongoing research is therefore addressing these challenges – for example, by employing techniques for explainable AI, federated and privacy-preserving learning, and rigorous validation – to ensure that deep learning on non-imaging clinical data is both effective and clinically acceptable.

In terms of imaging, early systems like MYCIN paved the way, but contemporary DL models, particularly Convolutional Neural Networks (CNNs), now dominate medical image analysis [15]. CNNs, has achieved state-of-the-art performance in image classification, object detection, and segmentation tasks. Transfer learning, where models pretrained on large datasets (e.g., ImageNet) are fine-tuned for medical imaging, enables effective learning with smaller clinical datasets.

2.2.1 Convolutional Neural Network (CNN)

A convolutional neural network (CNN) is a deep learning model composed of multiple stacked layers – primarily convolutional layers, pooling layers, and usually ending with fully connected layers [16]. In each convolutional layer, small matrices called kernels (filters) slide across the input image to produce feature maps that highlight specific patterns (e.g. edges, textures) wherever they appear [16]. Nonlinear activations (e.g. ReLU) are applied to these feature maps, and pooling operations (such as max-pooling) then downsample the data, reducing dimensionality while making the detected features more invariant to scale or position changes. By stacking multiple convolution and pooling layers, CNNs learn a hierarchy of features: early layers detect low-level details (boundaries, corners), while deeper layers combine these into higher-level features or shapes relevant to the task [16]. Inally, one or more fully connected layers integrate the extracted features to produce the output (for example, classifying an OCT scan as normal vs. diseased) [16]. The learnable weights of the CNN (the filter values) are optimized during training so that the network automatically learns the most useful features for the prediction task, rather than relying on manually engineered features [17].

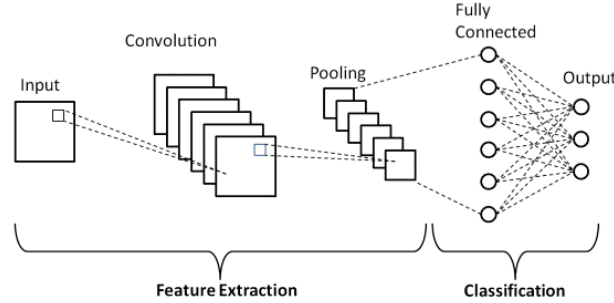


Figure 2.1: CNN Architecture [18]

CNNs are particularly effective for analyzing medical images like OCT because they can automatically recognize subtle, complex patterns in the data without explicit feature design [17]. The convolutional layers inherently confer translational invariance, meaning a feature learned (for example, a fluid pocket or abnormal layer in an OCT) can be detected anywhere in the scan [16]. Moreover, OCT scans have consistent imaging protocols (centered on the macula with similar orientation), which simplifies the computer vision task and allows CNN models to achieve high performance even with relatively small clinical datasets [19, 20]. Modern CNN architectures (e.g. ResNet and DenseNet) introduce innovations like skip connections and dense layer connectivity that enable training of much deeper networks, further improving accuracy in image recognition tasks [16]. These advanced CNN models (discussed in later sections) form the backbone of many current ophthalmology AI systems, successfully identifying diseases and features in OCT images to assist clinical decision-making.

2.2.2 Deep Learning in Ophthalmology and OCT Imaging

As discussed in the previous section about how AI and deep learning are influencing the medical field, Ophthalmology also suited for AI integration as there are high volume and standardization of imaging data associated with it. Optical Coherence Tomography (OCT) imaging provides high-resolution, cross-sectional views of the retina, which are essential for monitoring and diagnosing and monitoring retinal diseases such as AMD. The applications of AI in OCT imaging focus on prediction of disease trajectory and also on fluid quantification, layer segmentation. Deep learning models have shown promising results in analyzing OCT data to detect biomarkers, quantify retinal fluid, and predict disease progression.

In the paper [21], automated fluid detection is demonstrated using a CNN with U-Net architecture, where sensitivity and specificity of over 90% were achieved. Similarly, Mishra et al. (2020)[22] utilized shortest-path algorithms combined with CNNs for accurate segmentation of retinal layers and drusen in AMD. Beyond segmentation, AI models quantify biomarkers like subretinal fluid (SRF),

intraretinal fluid (IRF), and pigment epithelial detachments (PED), all of which are critical in managing wet AMD (Mares et al., 2024)[23].

AI algorithms also assist in predicting functional outcomes such as BCVA. Models trained on longitudinal OCT data can forecast disease activity and treatment needs, improving personalized care and resource allocation [24, 25].

In the context of wet-AMD, managing disease progression, predicting visual acuity, and determining treatment needs are critical. The target of this thesis is OCT-Based Classification of Visual Acuity in Patients Affected by Wet AMD through leveraging deep learning. The goal is to classify patients based on BCVA thresholds and treatment provided to them over a long period (between 7 to 14 years), determining whether they improved or not. Models like ResNet (Residual Networks) and DenseNet (Densely Connected Convolutional Networks) have proven successful in different literatures.

ResNET

He et al. (2015) [26] introduced this model in "Deep Residual Learning for Image Recognition" (CVPR), which uses residual skip connections that allow the model to learn identity mappings, enabling very deep networks to train without vanishing gradients. The advantages of ResNet include its robustness, as it is widely adopted in medical imaging. Versions like ResNet18/34 that are lightweight, work well on small datasets (e.g., OCT scans), are fast to train, less prone to overfitting with limited data, and demonstrate strong generalization in early detection tasks. De Fauw et al. (Nature Medicine, 2018) [24] used a ResNet-style backbone in their multi-disease OCT diagnosis pipeline.

DenseNET

Similarly, DenseNet (Densely Connected Convolutional Networks) has been used by many researchers working on AMD, such as for automated detection of fluid in wet AMD using DenseNet variants [27] and SRF/IRF quantification from swept-source OCT (using DenseNet-121). This model originated from Huang et al. (2017) [28] — "Densely Connected Convolutional Networks" (CVPR). In this architecture, every layer connects to all previous layers, promoting feature reuse and reducing parameter count while maintaining high performance. Some advantages of DenseNet include feature-richness, efficient parameter usage, and encouragement of gradient flow. It has proven to perform slightly better than ResNet on high-resolution B-scans with detailed biomarkers.

2.3 Deep learning for early diagnosis and treatment planning in Wet AMD

The way we treat wet AMD has changed dramatically with anti-VEGF injections. These shots help stop abnormal blood vessel growth and fluid leaks in the

eye. While they work well, patients often need many injections and check-ups, which can be tough to keep up with [8]. That’s why doctors want to predict early on how well a patient will see and how many injections they’ll need. OCT scans are the main tool doctors use to check wet AMD. They show tiny fluid pockets (IRF, SRF, PED) that tell us if the disease is active or improving. AI systems, especially CNN models, are now very good at measuring these fluid changes [21, 29]. These measurements help predict future vision and treatment needs.

Other eye scans like FAF, OCTA, and color photos add helpful information too. FAF shows damage to the back of the eye, while OCTA reveals blood vessel problems without needing dye. When combined with OCT, these give AI systems a more complete picture [30]. Recent research shows that using all these scans together with vision test results gives the most accurate predictions [31, 32].

It is useful for AI models to track how the eye changes over time, especially in diseases like wet AMD where progression and response to treatment evolve gradually. Unlike traditional models that analyze single timepoints, specialized temporal models such as Recurrent Neural Networks (RNNs) and Transformers are capable of capturing sequential dependencies in longitudinal OCT data. RNN variants like LSTM (Long Short-Term Memory) process OCT scans from multiple follow-up visits as a time series, allowing the model to learn the correlation between changes in retinal structure and visual outcomes [33, 34, 35]. Transformers are based on attention mechanisms rather than recurrence; they can weigh the importance of each time point with others, making them particularly effective for modeling long-term disease trajectories [36, ?]. These models have been increasingly applied in healthcare settings for tasks like disease progression modeling, future outcome prediction, and treatment recommendation [37, 38]. The learnings from temporal patterns help models to predict future treatment needs or visual acuity trends, allowing more proactive and personalized care planning [29].

This thesis focuses on using deep learning to perform static and temporal classification of visual acuity outcomes in patients with wAMD, based on longitudinal OCT imaging data. The static model is classifying each year’s B-scans independently as representing good or poor visual acuity, while the temporal model analyzes year-over-year progression to assess whether a patient’s condition has improved or not. The goal is to support doctors in evaluating treatment effectiveness and monitoring disease progression over time.

2.4 Explainability and Clinical Integration of AI in Ophthalmology

Deep learning models are criticized for their opacity despite their predictive accuracy, leading to limited trust in clinical decision-making—referring them

as the "black box". The field of Explainable Artificial Intelligence (XAI) has introduced techniques that are designed to interpret model behavior in order to highlight features influencing specific predictions. Among these techniques, Gradient-weighted Class Activation Mapping (Grad-CAM) is used widely to generate heatmaps that visualize salient regions in the input image that most impact the output decision. Even in ophthalmology, Grad-CAM successfully applied to OCT scans and this help in localizing disease-relevant regions, such as areas with fluid accumulation or retinal layer disruption. For example, Karri et al. (2017) [39] used deep learning models with visualization techniques like Grad-CAM to identify pathological biomarkers in retinal OCT images, which provides model focus understanding and clinical interpretability enhancement. Similarly, De Fauw et al. (2018) [24] also used visual explanation frameworks to justify referral decisions in retinal disease triage systems, which help in bridging the gap between algorithmic predictions and clinical insight.

Beyond just showing results, tools like SHAP help explain how AI models make decisions by showing which factors most influence their predictions. This is especially helpful when combining medical records and scan data. For AI to be safely used in real clinics, it needs to work well for all kinds of patients - not just those in initial tests. Big databases like IRIS and studies like PINNACLE collect information from many different hospitals to test if these AI tools work consistently everywhere. Some rules are also in process of creation by Government health agencies (like the FDA) to make sure medical AI is safe, accurate, and easy to understand before doctors can use it, this makes clear explanations of how the AI works very important.

2.5 Challenges and Future Directions in AI for Wet AMD

Although the progress in applying artificial intelligence (AI) to retinal diseases is considerable, still the development of reliable AI models for Wet Age-Related Macular Degeneration (Wet AMD) faces several key challenges. One of the primary obstacles is the heterogeneity of OCT data. Differences in imaging resolution, device manufacturers, and scanning protocols across clinics can lead to domain shifts that degrade model performance and limit generalizability [24]. Moreover, public datasets specifically tailored to Wet AMD are scarce, and many existing studies rely on proprietary data, making external validation and benchmarking difficult [21, 40].

A major challenge in Wet AMD modeling is the temporal nature of disease progression. Most AI models to date perform snapshot-based analysis, ignoring the sequence of anatomical and functional changes over time. However, real-world Wet AMD management involves tracking of changes over months or years to assess response to anti-VEGF therapy. Recent research suggests that integrating longitudinal data can enhance predictive accuracy [31], but designing

architectures that learn effectively from such sequences remains a non-trivial task.

Another critical area in this domain is explainability. Clinicians are hesitant to adopt AI systems unless they understand the reasoning behind the predictions. Although techniques like Grad-CAM or Grad-CAM++ can highlight regions of interest [41], these maps do not yet offer the level of clinically grounded interpretability necessary for medical decision-making [42].

Multi-task learning frameworks offer a promising direction jointly predict BCVA, number of injections, and disease trajectory [43]. Lastly, AI for Wet AMD progression will require close collaboration between AI researchers, and clinicians, along with standardized datasets and evaluation metrics to enable scalable, trustworthy deployment in clinical practice.

Chapter 3

Data and Preprocessing

The foundation of any robust deep learning system lies in the quality and organization of its input data. This chapter details the acquisition, curation, and preprocessing pipeline for the longitudinal ophthalmic dataset used in this study.

3.1 Data Source and Collection

The dataset used in this thesis was given by the Ospedale Sacro Cuore – Don Calabria, located in Negrar, Italy. It consists of longitudinal ophthalmic records of 325 eyes from 271 patients diagnosed with wet Age-related Macular Degeneration (wAMD). These patients were followed over a period ranging from 8 to 14 years, with a mean follow-up duration of 9.5 years.

The original scan data was available only through the proprietary Heidelberg Engineering software in the ‘e2e’ format, which encapsulates 3D volumetric OCT data along with fundus images and metadata. Due to the constraints in the exportation system, data could not be retrieved in standard formats such as DICOM or PNG directly in bulk. As the number of patients were larger, manual downloading of data was not feasible. To overcome this, the open-source Python library OCT-Converter¹ was used [44].

This utility allowed for the extraction of individual 2D B-scans in PNG format and associated metadata in JSON format from the ‘e2e’ files. Despite high success, 4 out of 325 eyes were not converted due to file corruption or unexpected format inconsistencies, leaving 321 eyes available for subsequent analysis.

¹<https://github.com/marksgraham/OCT-Converter>

3.2 Dataset Organization

The dataset which got converted from .e2e to .png files was organized in a hierarchical directory structure to enable effective traceability and downstream processing. Each patient was assigned a top-level folder named in the format `SURNAME, FIRSTNAME`. Inside each patient’s folder, separate subfolders contain follow-up sessions, identified by a unique follow-up session name such as `ACCOR01L`.

- MainFolder: `SURNAME, FIRSTNAME`
 - Subfolder: `ACCOR01L`
 - Subfolder: `ACCOR02L`

The reasoning behind the unique follow-up session name like ‘ACCOR01L’ is as follows: ‘ACCOR’ represents the patient’s last name, ‘01’ represents the follow-up number (indicating it’s the first follow-up), and ‘L’ is the first letter of the patient’s first name.

Each subfolder contains:

- B-scan images in .png format
- `Annotations.json`: Full metadata from the original .e2e extraction
- `FilteredAnnotations.json`: A curated metadata subset relevant to this study

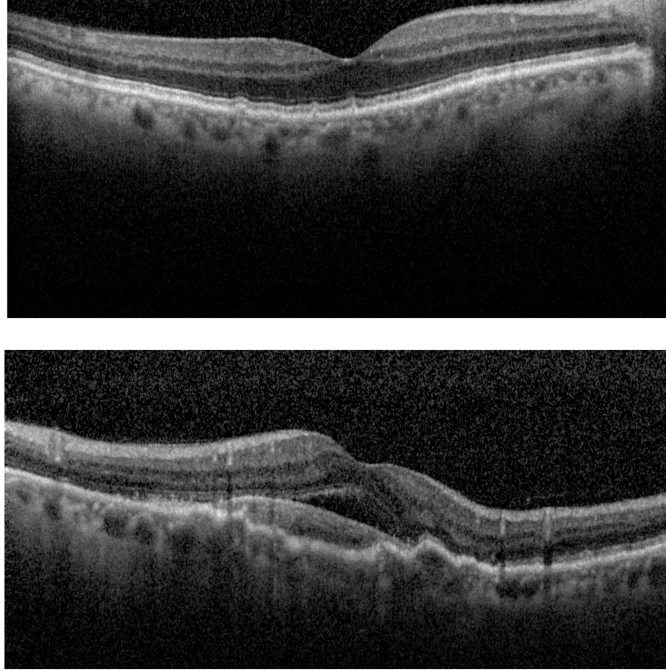


Figure 3.1: B-scan images of both eyes - Right eye (upper image) and Left eye (lower image)

This structure supports efficient access to both imaging and metadata components required for the modeling tasks.

3.2.1 Metadata

The metadata extracted alongside each B-scan includes fields such as:

- `filename`
- `Laterality`
- `scan_date`
- `imgSizeX`
- `imgSizeY`
- `imgQuality`
- `acquisitionTime`

The `Laterality` tag indicates whether the scan corresponds to the left or right eye, and `scan_date` provides temporal context for longitudinal analysis.

3.2.2 Clinical Data Integration

A structured Excel sheet provided by the hospital contained annual clinical data per eye. Key fields included:

- Patient identifiers: `AACognome`, `Nome`, `OCCHIO`
- Visual acuity metrics: `BCVA baseline`, `BCVA 1 anno` to `BCVA 12 anno`
- Treatment history: `N. inj 1 anno` to `N. inj 12 anno`

Each patient-eye record from the Excel sheet was linked to the corresponding folder in the image dataset. This mapping allowed the extraction of longitudinal labels (e.g., visual outcome, injection frequency) for supervised learning.

3.3 Preprocessing

After the dataset organization, the next step is of preprocessing, where the filtering, eye separation, resizing, and coloring of the images are done.

3.3.1 Image Filtering

To ensure data consistency, filtering was essential. The following criteria were used for filtering:

- Scans with missing or invalid metadata (e.g., empty or unparsable `scan_date`) were removed
- Files without `Laterality` information were excluded
- Images with abnormal dimensions (not 512×496) or corrupted pixels were discarded

This preprocessing ensured that only high-quality, well-annotated scans were retained for modeling.

3.3.2 Eye Separation and Matching

Each `.e2e` file could contain both left and right eye scans. The `Laterality` field in the metadata `L` or `R` was used to associate scans with the appropriate eye correctly. This was crucial for matching clinical records with imaging data. Any images lacking laterality were excluded from further analysis to prevent eye-wise mislabeling.

3.3.3 Resizing and Coloring

Following the filtering and eye separation, the valid B-scan images were converted to **gray scale** to reduce input complexity and resized to **224x224** pixels. This resizing step standardizes image dimensions across all samples, making them compatible with common convolutional neural network architectures. The grayscale transformation also minimizes computational cost without sacrificing diagnostic features.

3.3.4 Yearly Grouping of Follow-Ups

As the imaging data contained bi-monthly follow-ups, but the clinical Excel data was only available annually, a grouping mechanism was employed. Therefore, the B-scans were **clustered by year** as follows:

- **Year 1:** baseline scan + follow-ups within **30 days** of the baseline.
- **Years 2–12:** subsequent scans grouped in **360-day intervals** after baseline.

This allowed monthly scans to be grouped and assigned to annual clinical labels for consistent temporal modeling.

3.4 Labeling Strategy

As the objective is to tackle the two related but distinct classification tasks on longitudinal OCT data for wet AMD patients: static classification and temporal classification. In this section, the labelling strategies for static and temporal models are discussed.

3.4.1 Static Labeling Strategy

For a time-independent classification model, each year’s scan group was labeled based on the BCVA value:

$$\text{Label}_{\text{year}} = \begin{cases} \text{good}, & \text{if } \text{BCVA}_{\text{year}} \geq 0.4 \\ \text{poor}, & \text{otherwise} \end{cases}$$

This binary classification task focused on predicting visual quality without considering prior history.

3.4.2 Temporal Labeling Strategy

For temporal modeling, labels were assigned based on year-over-year change in BCVA (ΔBCVA).

$$\Delta\text{BCVA}_n = \text{BCVA}_n - \text{BCVA}_{n-1}$$

If the improvement between the current and the most recent prior year with valid data was ≥ 0.1 , the year was labeled as ‘improved’, otherwise ‘not improved’. Year 1 was always labeled as ‘baseline’.

$$\text{Label}_n = \begin{cases} \text{"baseline"}, & \text{if } n = 1 \\ \text{"improved"}, & \text{if } \Delta\text{BCVA}_n \geq 0.1 \\ \text{"not improved"}, & \text{if } \Delta\text{BCVA}_n < 0.1 \end{cases}$$

This enabled the use of RNN/LSTM architectures to learn disease progression patterns.

3.5 Annotation Format Example

As of now, we know how the clinical data is integrated, how yearly grouping of follow-ups is done, and how the labelling is working, so each final annotation entry follows a JSON format, which is shown below:

```
{
  'year': 2,
  'images': [...],
  'BCVA': 0.2,
  'n_inj': 6,
  'label': 'not_improved',
  'scan_date': '22/10/2009'
}
```

These per-year groupings per patient-eye were used as input for both static and sequential models.

3.6 Challenges and Resolutions

In this process, several technical hurdles were encountered:

- `.e2e` files required specialized handling as there is no native `DICOM/PNG` output for bulk.
- Eye separation was non-trivial and relied on the `Laterality` tag.
- Some follow-ups had missing or ambiguous metadata, requiring fallback strategies based on folder order and timestamps.
- Matching clinical Excel entries to image folders required robust normalization of patient names and handling of inconsistencies.

Despite these challenges, a comprehensive and clean dataset was curated with strong clinical relevance and temporal depth.

Chapter 4

Methodology

In this chapter, we will talk about our objective in detail, which is to perform static and temporal classification on longitudinal OCT data. Furthermore, dive deeper into the model architectures used for both of the classification tasks, from discussing their basic architecture to the implementation details and training. We will also cover the interpretability of the model with Grad-CAM.

4.1 Classification Objectives: Static vs. Temporal Classification

In this thesis, two related but distinct classification tasks on longitudinal OCT data for wet AMD patients are tackled: static classification and temporal classification. In the static setting, the goal is to analyze OCT scans (from a specific year) and classify the patient’s visual acuity status for that year in isolation (e.g. “good” vs. “poor” vision) without considering any prior or future scans. In contrast, the temporal classification task leverages a sequence of OCT scans from successive follow-up years to predict changes in visual acuity over time. Essentially, the temporal model learns from the year-over-year progression: given a series of past scans, it predicts whether the patient’s condition improved or not improved in the next year (based on changes in Best Corrected Visual Acuity, BCVA). Thus, while static classification focuses on snapshot outcomes at individual timepoints, temporal classification focuses on longitudinal outcomes and trend prediction across time. This dual approach allows us to assess both cross-sectional visual acuity status (static context) and disease progression (temporal context) for wet AMD patients.

4.2 Model Architectures for Static Classification

For the static OCT classification, three deep convolutional neural network (CNN) architectures are experimented: **ResNet18**, **ResNet34**, and **DenseNet121**. These architectures were chosen based on their proven success in medical imaging literature and their balance of depth vs. complexity [26]. ResNet (Residual Network) models introduce identity skip connections that help train very deep networks without vanishing gradients [26]. ResNet18 and ResNet34 are selected as they are relatively lightweight (18-layer and 34-layer variants) and have shown robustness on smaller datasets like OCT scans [26]. DenseNet (Densely Connected Network), on the other hand, features a pattern of **dense connectivity** where each layer feeds its output to all subsequent layers [28]. DenseNet121 (with 121 layers) was included for its ability to encourage feature reuse and efficient parameter utilization, which can yield richer representations and slightly better performance on detailed OCT images [28]. By comparing ResNet and DenseNet families, it was aimed to understand the impact of network depth and connectivity pattern on OCT-based visual acuity classification.

Adaptation to OCT Data (Grayscale Inputs). A key consideration was that our OCT B-scans are single-channel (grayscale) images, whereas these CNN architectures are typically designed for 3-channel RGB images. Each network’s input layer modified to accommodate 1-channel input. In practice, this meant **replacing the first convolution layer** (originally expecting 3 channels) with a kernel of identical size but with a single input channel. The new 1-channel filters are initialized by averaging the pre-trained weights across the RGB channels, effectively preserving the learned features in a grayscale-compatible way. This approach allowed us to leverage transfer learning from ImageNet – the models were initialized with weights pre-trained on a large dataset (ImageNet), then fine-tuned on our OCT data. Using such pretrained models helped given our limited medical imaging data, as prior works have shown improved learning and convergence when fine-tuning ImageNet CNNs on medical images.

4.2.1 ResNet18 and ResNet34

ResNet18 and ResNet34 are CNN architectures that utilize **residual learning** to facilitate training deeper networks [26]. In a ResNet, layers are organized into groups of residual blocks Figure 4.1 where each block’s output is added to its input via an identity shortcut connection. This means the block actually learns a residual function $F(x) = H(x) - x$, so that the block’s output is $H(x) = F(x) + x$. By learning modifications to the identity rather than entirely new mappings, ResNets avoid the degradation problem and can be substantially deeper without loss of performance [26]. The **ResNet18** model consists of 8 such basic residual blocks (2 per each of four stages), while **ResNet34** uses a deeper stack of 16 blocks (4 per stage), effectively doubling the depth. Apart from depth, ResNet18/34 share a similar architecture: an initial 7×7 convolution and max-pool, followed by four stages of convolutional blocks (with 64, 128, 256, 512

filters respectively), and finally a global average pooling and fully-connected layer.

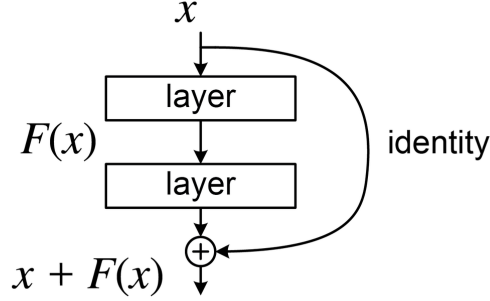


Figure 4.1: A **residual block** used in ResNet architecture. The input x is forwarded (identity shortcut) and added to the output of a sequence of layers computing $F(x)$ (e.g., Conv+BN+ReLU). This skip connection enables the network to learn residual functions ($F(x)$) on top of the identity, which greatly eases training of deep networks by mitigating vanishing gradients. ResNet34 stacks such residual blocks (with occasional downsampling layers) to form a deep CNN for image classification [26].

The ResNet architectures are customized for our binary classification task as follows. First, the conv1 layer was changed to accept 1-channel input (as described above). Second, the final fully-connected layer (originally 1000-neuron for ImageNet) was replaced with a single-output neuron to predict the probability of “good visual acuity” (since we framed static acuity prediction as a binary classification: good vs. poor). We kept the majority of the ResNet layers as a feature extractor and only changed this classifier layer. In practice, the modified ResNet’s forward pass can be described in stages:

- **Feature Extraction:** The input OCT image passes through the convolutional backbone (conv layers + residual blocks), yielding a high-level feature map from the last residual block. For ResNet34, this feature map has 512 channels (after the layer4 block).
- **Global Pooling & Flattening:** Global average pooling applied to this feature map, converting it into a 512-dimensional feature vector. (This pooling effectively “flattens” the spatial dimensions by averaging, so no explicit flatten operation is needed beyond collecting the pooled values.)
- **Classification Head:** The 512-d feature vector is fed into a fully connected layer of size $512 \rightarrow 1$ (one output neuron). This linear layer produces a logit score for the positive class (e.g., probability of good acuity).
- **Sigmoid Activation:** A sigmoid function $\sigma(z)$ is applied to the output logit to obtain a probability in $[0, 1]$ for the positive class. We prefer Sigmoid because it naturally suits binary classification by outputting a probability of

the single class being “true”. (During training, we used a binary cross-entropy loss with this sigmoid output.)

Dropout regularization is also introduced in the classifier to mitigate overfitting. Specifically, for ResNet34 a dropout layer (with $p = 0.5$) was inserted right before the final fully-connected layer. This means during training, 50% of the features from the global pool were randomly zeroed out, forcing the model to not overly rely on any one feature. This strategy improved generalization given the limited size of our dataset. Overall, ResNet18/34 provided strong baseline models that are fast to train and less prone to overfitting on smaller data [26].

4.2.2 DenseNet121

DenseNet121 is a deep CNN that further extends the idea of feature reuse introduced by ResNet. Instead of summing identity mappings, DenseNet concatenates outputs from earlier layers to later layers [45]. Figure 4.2 illustrates the internal structure of a single dense block, a key component of the DenseNet architecture which is typically organized into four such blocks separated by transition layers [46]. In each dense block, there are several sequential dense units (convolutional layers typically consisting of BatchNorm + ReL + 3×3 Conv). What makes it “dense” is that each dense unit receives as input the feature-maps of all previous units in that block (the inputs are concatenated along the channel dimension) [45]. If a dense block has N layers, the n -th layer gets feature-maps from layers $1, 2, \dots, (n - 1)$ as additional inputs. This connectivity yields $\frac{N(N+1)}{2}$ direct connections in a block [45] – an extreme case of skip connections. As a result, DenseNet layers can directly access low-level features from early layers, encouraging feature reuse and efficient propagation of gradients [28]. The “121” in DenseNet121 refers to the total number of layers (including conv layers inside dense units and transition layers).

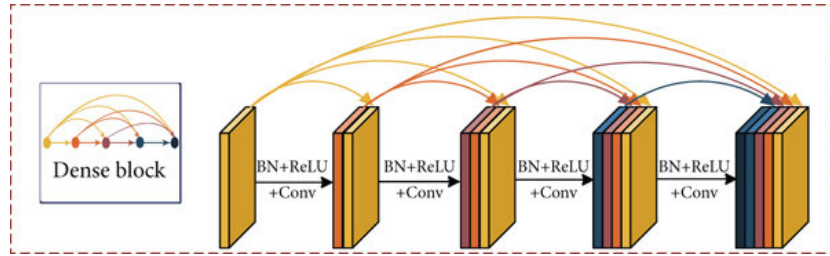


Figure 4.2: **DenseNet architecture** and dense block connectivity. Each dense block (yellow box) consists of multiple convolutional layers (dense units) where each layer takes **all preceding feature-maps as input** (illustrated by the multiple arrows). The feature-maps are concatenated channel-wise before feeding into each new layer’s BN–ReLU–Conv operations. This dense connectivity pattern enables maximum feature reuse and strengthens gradient flow through the network [46].

The DenseNet121 architecture starts with an initial convolution (7×7) and pooling, similar to ResNet, then has: Dense Block 1 (with 6 conv layers), transition layer, Dense Block 2 (12 conv layers), transition, Dense Block 3 (24 conv layers), transition, Dense Block 4 (16 conv layers), and finally a global average pool and fully-connected layer. Each dense block’s layers output a fixed number of new feature maps (called the growth rate, e.g. $k = 32$), and these get concatenated to the running total of features. By the end of the last dense block, the feature vector is very high-dimensional (around 1024 channels for DenseNet121 after concatenation). A transition layer (which consists of a 1×1 convolution followed by 2×2 average pooling) is used between blocks to down-sample and compress the feature size, preventing the network from blowing up in complexity [46].

For our implementation, DenseNet121 is adapted in a similar fashion as ResNet: the first conv layer was modified to 1-channel input, and the final classifier was replaced. After the final dense block, DenseNet has a global average pooling that produces a 1024-dimensional vector. We fed this through a new fully-connected layer to produce a single output logit, followed by a sigmoid activation for the probability of the positive class. We also added a dropout ($p = 0.5$) before this final layer to regularize the model. Notably, we chose DenseNet for its potential advantages highlighted in prior studies – **feature-rich representations and improved gradient flow** – which can be beneficial for capturing subtle structures in OCT scans [28]. DenseNet’s propensity for feature reuse means even fine details (like small fluid pockets or lesions) detected by early layers could be carried through and influence the final prediction, an appealing property for OCT analysis. In our static classification results (see Chapter 5), DenseNet121 indeed outperformed the ResNet models, which motivated using DenseNet as the basis for our temporal model.

4.2.3 Implementation Details and Training

All models were implemented in PyTorch. The pretrained versions of ResNet18, ResNet34, and DenseNet121 are utilized from torchvision.models (ImageNet weights) and then made the above modifications (changing input conv layers and replacing classifier layers). Model weights were fine-tuned on our OCT dataset using binary cross-entropy loss. The Adam optimizer is used for training, with an initial learning rate of $1e-4$. Each training image was augmented with slight random rotations, flips, and intensity normalization to improve robustness. It is ensured that for static classification, training batches contained images from multiple patients and years to help the model generalize beyond patient-specific features.

4.3 Temporal Classification Using ResNet+LSTM

Given the lightweight model, ResNET is adopted as the visual feature extractor for the **temporal model**. The temporal modeling aims to classify whether a

patient’s BCVA **improved or not** in each subsequent year, by looking at a sequence of that patient’s past OCT scans. Recurrent neural networks are well-suited for such sequential data, and an **LSTM** (Long Short-Term Memory) network is chosen for modeling the longitudinal dependencies [33, 34, 35]. By combining a CNN (for spatial feature extraction from images) with an LSTM (for temporal sequence learning), our model can learn how changes in retinal morphology over time relate to changes in visual acuity.

4.3.1 Architecture Design (ResNet18 + LSTM)

The hybrid model is structured as follows:

- **Per-Scan Feature Encoding:** B-scans for a given year are passed through the ResNET18 network (up to the global pooling layer, excluding its classifier). This converts the image into a 512-dimensional feature vector f_t for year t . These ResNet weights are shared for all time steps (the same CNN processes each year’s image). The result is a sequence of feature vectors $[f_1, f_2, \dots, f_T]$ for a patient with T years of follow-up.
- **Sequence Formation:** We sort the scans of each patient chronologically. Since year 1 is defined as the baseline (diagnosis time), we do not predict an improvement label for year 1 (there is no prior year to compare). For each subsequent year n (year 2, 3, ...), we take all available feature vectors up to year $n-1$ as the input sequence and the label indicating improvement at year n as the target. Here, ‘improved’ vs. ‘not improved’ for year n is determined by the change in BCVA from year $n-1$ to n (as described in Chapter 3, if $\Delta BCVA_n \geq 0.1$ the year is labeled improved). For example:
 - To predict the outcome at Year 2, the model is given the sequence of features $[f_1]$ (year 1 only).
 - To predict the improvement in year 3, the model is given $[f_1, f_2]$ (years 1 and 2).
 - To predict Year 4, it receives $[f_1, f_2, f_3]$, and so on.

Each of these constitutes a training sample for the LSTM. Thus, a patient with say 5 years of scans yields 4 training sequences: (years [1] -> label2), (years [1,2] -> label3), (years [1,2,3] -> label4), (years [1,2,3,4] -> label5).

- **LSTM Network:** The sequence of ResNet feature vectors is then fed into an LSTM to model the temporal dynamics. Our LSTM was configured with hidden size 128 and 2 layers (stacked LSTM), operating in batch-first mode (input shape batch $\times$ seq $\times$ feature). We used a unidirectional LSTM (since we only progress forward in time). The LSTM processes the feature sequence and produces an output at each time step. We are interested in the output at the final time step of the sequence, which encodes information from all prior scans.

- **Temporal Classifier:** The last LSTM output (a 128-d vector) is passed through a fully connected layer to produce a single logit for the “improved” class at the next year. A sigmoid activation is applied to convert this logit into a probability of improvement. This is then compared to the true label (“improved” or “not improved”) for that year during training.

In summary, the ResNet+LSTM forms an end-to-end trainable pipeline: ResNet extracts high-level imaging features for each visit, and the LSTM learns how those feature patterns evolve in patients who improve vs. those who do not. The model effectively learns a function $F([f_1, \dots, f_{n-1}]) \rightarrow \text{Prob}(\text{improved at year } n)$.

Training the sequence model required handling patients with varying numbers of follow-ups. Shorter sequences are padded with a special <PAD> token (zero-vector features) and used PyTorch’s packed sequence utilities so that padded values don’t affect LSTM computation. The LSTM model is trained using binary cross-entropy loss applied at each predicted year outcome. Notably, because patients contribute multiple training samples (one for each follow-up year beyond the first), it was taken care to use a patient-wise split for training/validation to avoid leaking information (ensuring that all sequences from a given patient are entirely in either training or validation set). The LSTM was trained with the Adam optimizer (learning rate 1e-3) and a dropout of 0.3 on the LSTM’s recurrent connections for regularization.

By using ResNet features, the LSTM benefits from a rich representation of each scan, and by using an LSTM, the model can capture patterns like “if the baseline scan had certain severe pathology and the follow-up scans show X change, the outcome tends to be improvement”. This sequential modeling approach is similar to how a clinician would compare OCT scans over visits to gauge whether the patient is responding to treatment or declining. Leveraging LSTMs for such longitudinal medical imaging data is supported by recent studies, which show RNNs can effectively learn progression trends in diseases like AMD. Our ResNet-LSTM is a novel application in the context of predicting visual acuity outcomes across multiple years. We also plan to integrate an attention mechanism to help the model weight the importance of each time step, and to concatenate relevant clinical metadata (e.g., number of injections) with the image features, providing the LSTM with clinical context.

4.4 Model Interpretability with Grad-CAM

While deep learning models can achieve high accuracy, their black-box nature is a concern in clinical settings. To build trust and provide insights into the model’s decisions, we incorporated an explainability technique: Gradient-weighted Class Activation Mapping (Grad-CAM). Grad-CAM generates a heatmap over the input image, highlighting the regions that were most influential in the model’s prediction. This helps us verify whether the model is looking at clinically rele-

vant features (such as lesions, fluid pockets, etc.) in the OCT when predicting visual acuity.

Grad-CAM works by using the gradients of the model’s output with respect to feature maps of a deep convolutional layer to determine the importance of each neuron for the decision [47]. For our case, Grad-CAM is applied to the model. The final convolutional layer is chosen as the target for Grad-CAM, since those feature maps have the highest-level information and still retain spatial layout. The process is as follows:

- **Forward Pass & Feature Extraction:** A grayscale OCT B-scan is given as input into the trained model and obtain the prediction (sigmoid output). During this forward pass, intercepted and saved the activation maps A^k from the final convolutional layer. These activation maps have a spatial resolution (downsampled relative to the input, e.g. 7×7 or 14×14) and multiple channels (let’s say k indexes the channel).
- **Backpropagate Gradients:** Then computed the gradient of the predicted class score (for example, the logit for “good vision” if the model predicted good) with respect to the feature maps A^k . This involves performing a backward pass from the output to that layer. Denoted these gradients as $\frac{\partial y}{\partial A^k}$.
- **Global Pooling of Gradients:** The gradients flowing back are spatial (same size as A^k). Grad-CAM computes a channel-wise importance weight by global average pooling the gradient across the spatial dimensions [47]. In other words, for each feature map k calculated is:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k},$$

where i, j index the pixels of the feature map and Z is the total number of pixels. The weight α_k reflects how important feature k was for the target output y (positive values indicate that increasing that feature increases the output).

- **Weighted Combination:** Next, the saved activation maps are combined using these weights:

$$L_{\text{Grad-CAM}}(x, y) = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right),$$

where we take a weighted sum of the feature maps A^k and then apply a ReLU. The ReLU ensures that only features that have a positive influence on the class are considered (negative influences are set to 0) [47]. The result is a single 2D map (let’s call it L) which is essentially a coarse heatmap of the same spatial size as the feature maps.

- **Upsampling to Image & Overlay:** Then upsample this low-resolution heatmap L to the size of the original OCT image (using bilinear interpolation). The heatmap values are normalized to $[0,1]$ and then mapped to a color map (e.g. jet colormap) for visualization. Finally, we overlay this colored heatmap on the OCT scan to highlight the important regions.

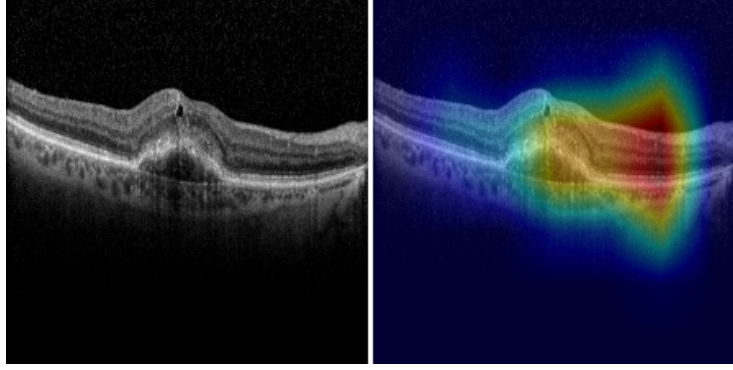


Figure 4.3: Grad-CAM Visualization

In our implementation, we automated this process by registering PyTorch hooks on model’s selected layer output. The forward hook captures the feature maps, and the backward hook captures gradients when we call `.backward()` on the output score. Then the above calculations is performed to get the Grad-CAM heatmap. These heatmaps are generated for numerous test scans to interpret the model’s focus. An example Grad-CAM visualization is shown in Figure 4.3, which depicts an OCT B-scan with a superimposed heatmap indicating the regions most influential for predicting poor visual acuity. In many cases, Grad-CAM highlighted central foveal lesions or large fluid areas, which aligns with clinical intuition that these features impact vision. This added interpretability is crucial – even if the model predicted correctly, we want to ensure it did so for the right reasons. Grad-CAM provides a qualitative check: if the model were focusing on irrelevant regions (like image corners or noise), that would indicate a problem; instead, seeing the heatmaps align with disease pathology gives confidence in the model’s reliability.

In summary, by using Grad-CAM we bridged the gap between the black-box CNN decisions and human-understandable visual cues.

Chapter 5

Experiments and Results

In this chapter, the experimental setup and results are presented for our two-pronged approach: static classification and temporal classification of visual acuity in wet AMD patients. The static analysis evaluates models that predict a patient’s visual acuity category (good vs. poor) from OCT scans of a single year in isolation, whereas the temporal analysis evaluates models that predict year-over-year improvement in acuity by leveraging sequences of scans across multiple follow-up years. Results are reported under two data splitting strategies for the static task ‘image-wise’ (image-wise splitting divides the dataset based on individual OCT images, treating each scan independently) vs. ‘patient-wise’ (patient-wise splitting ensures that all images from the same patient (or eye) are assigned exclusively to either the training, validation, or test set. This approach is more clinically rigorous, as it prevents data leakage that could artificially inflate model performance) to examine generalization, and are evaluated using two prediction strategies for the temporal task (using the full history of past scans vs. using only the last-year scan) to assess the value of long-term context. All results are averaged across cross-validation folds for robustness.

Code Availability: All code used to implement the static and temporal classification experiments, including model architectures, data preprocessing, training routines, and evaluation scripts, is available at: ¹.

For static classification, we treat ‘**good vision**’ as the **positive class** (i.e., a true positive is a correctly identified good-vision exam). For temporal classification, we focus on the ‘**not improved**’ class as **positive** – in other words, a TP (true positive) indicates a correctly identified case of no improvement. We choose ‘not improved’ as the positive class in the temporal task because it is the prevalent outcome and of clinical interest to identify those who fail to improve.

¹<https://github.com/JawwadMehdi14/oct-visual-acuity-wet-amd>

To quantify model performance, a suite of standard classification metrics and visualization tools is used.

- **Accuracy:** the proportion of correct predictions over all samples (both “good” and “poor” acuity cases). This gives an overall success rate but can mask class imbalances.
- **Precision:** the fraction of positive predictions that were positive. In our binary context, if we define the “good vision” class as the positive class (for static classification) or “not-improved” as positive (for temporal classification), $\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$. High precision means when the model predicts improvement (or good acuity), it is usually correct, indicating a low false-alarm rate.
- **Recall (Sensitivity):** the fraction of actual positive cases that the model correctly identified: $\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$. High recall means the model catches most of the true ‘good’ vision cases in static classification and ‘not improved’ cases in temporal classification, with few missed. This shows that our model does not miss many bad vision cases in the static classification as the FN (false negatives) is close to zero for high recall. There is often a trade-off between precision and recall.
- **F₁-score:** the harmonic mean of precision and recall, $F_1 = 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$. The F₁ provides a single measure of balanced performance, especially useful under class imbalance.
- **AUC (Area Under the ROC Curve):** measures the classifier’s ability to discriminate between classes across all decision thresholds. An AUC of 1.0 indicates perfect class separation, while 0.5 indicates no better than chance. We report the AUC to complement the accuracy, as the AUC is insensitive to the choice of threshold and class distribution, reflecting the inherent discriminative power of the model.
- **Confusion Matrix:** a 2×2 table detailing the counts of TP (true positives), FP (false positives), TN (true negatives), and FN (false negatives). We will use confusion matrices to analyze the types of errors. For example, in a confusion matrix for the static task, the entries show how many yearly exams with good vs. poor acuity were correctly predicted or misclassified. This helps identify whether the model tends to overpredict one class (e.g., labeling poor-vision eyes as ‘good’ incorrectly or vice versa). In the (Appendix A), we provide representative confusion matrices to illustrate the error distribution of our best models.

All metrics are calculated per fold on the test portion and then reported as the mean (and standard deviation) across folds to account for variability. We perform 3-fold cross-validation in both scenarios. In image-wise CV, each fold’s test set is a random sample of individual exams (with potential patient overlap across folds), whereas in patient-wise CV, each fold’s test set consists of entire patients’ data, with no overlap of patients between folds. Next, we detail

the results for static and temporal classification experiments, including model architectures, training parameters, and performance comparisons.

5.1 Static Classification Results

For static OCT-based classification (predicting visual acuity category from a single year’s scans), we evaluated three deep CNN architectures: **ResNet18**, **ResNet34**, and **DenseNet121**. These models were chosen for their strong performance in medical imaging literature and were described in detail in Chapter 4. In brief, ResNet18 and ResNet34 are residual networks of 18 and 34 layers, and DenseNet121 is a 121-layer densely-connected network; all were initialized with ImageNet pretrained weights and adapted to single-channel OCT input (grayscale) as discussed in the Methodology chapter. We fine-tuned each model on our dataset using the **Adam optimizer** (initial learning rate 1×10^{-4}) and **binary cross-entropy loss** for the binary classification (“good” vs. “poor” vision). Each training was run for **30 epochs** per fold with a **mini-batch size** of **32**, using data augmentation (random flips, rotations, and intensity normalization) to improve robustness. Model selection was based on validation performance (we retained the epoch with the highest validation F_1 -score to avoid overfitting). We performed a **3-fold cross-validation** in all experiments: in each fold, 2/3 of the data were used for training and 1/3 held out for testing, rotating such that each sample appears in the test exactly once. To evaluate the impact of data leakage on model performance, we compared two split strategies: **(1) image-wise splitting**, where OCT images are randomly partitioned across folds, and **(2) patients’ eye level splitting**, which strictly segregates all images from a single patient’s eye into the same fold, ensuring clinically realistic validation.

Training Behavior: During training on the image-wise split, all models converged within roughly 15–20 epochs, after which validation loss plateaued. No explicit data augmentation was used for static classification; instead, the models relied on transfer learning from ImageNet-pretrained weights to improve generalization on limited OCT data. Validation metrics (accuracy, F1, AUC) closely tracked training metrics throughout training, suggesting minimal overfitting. The use of mixed-patient batch composition where each mini-batch included scans from different patients and years proved beneficial by preventing the network from memorizing patient-specific characteristics and encouraging learning of more generalizable features (e.g. structural biomarkers of disease severity). Early stopping with patience settings further helped avoid overfitting by halting training when validation performance plateaued.

Below, we present the results for each scenario, including per-fold performance and a comparative summary of the architectures.

5.1.1 Image-Wise Splitting Results

Under image-wise splitting, the training and test sets were randomly mixed at the level of individual yearly exams (while ensuring that all B-scans from a given exam stay together with their label). This means the models could see different follow-ups of the same patient in training and testing.

ResNet18 Results

The performance of Resnet18 (Appendix A.1.1) is summarized below in Table 5.1, with mean and standard deviation.

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.7818	0.8402	0.8283	0.8168 ± 0.0308
Precision	0.6953	0.7859	0.7658	0.7490 ± 0.0475
Recall	0.9174	0.8864	0.8914	0.8984 ± 0.0166
F ₁ -score	0.7911	0.8332	0.8238	0.8160 ± 0.0220
AUC	0.9097	0.9280	0.9280	0.9200 ± 0.0093

Table 5.1: ResNet18 performance across 3-fold cross-validation (mean $\pm$ standard deviation)

ResNet34 Results

The performance of Resnet34 (Appendix A.1.2) is summarized below in Table 5.2, with mean and standard deviation.

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.8318	0.7821	0.7696	0.7945 ± 0.0328
Precision	0.7852	0.6986	0.7063	0.7300 ± 0.0479
Recall	0.8624	0.9078	0.8361	0.8688 ± 0.0362
F ₁ -score	0.8220	0.7896	0.7657	0.7924 ± 0.0282
AUC	0.9182	0.9036	0.8680	0.8966 ± 0.0258

Table 5.2: ResNet34 performance across 3-fold cross-validation (mean $\pm$ standard deviation)

DenseNet121 Results

The performance of Resnet34 (Appendix A.1.3) is summarized below in Table 5.3, with mean and standard deviation.

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.8471	0.8346	0.8301	0.8372 ± 0.0088
Precision	0.8089	0.7842	0.7713	0.7881 ± 0.0190
Recall	0.8650	0.8729	0.8850	0.8742 ± 0.0100
F ₁ -score	0.8360	0.8262	0.8242	0.8287 ± 0.0062
AUC	0.9258	0.9234	0.9209	0.9234 ± 0.0024

Table 5.3: DenseNet121 performance across 3-fold cross-validation (mean $\pm$ standard deviation)

As expected, this scenario yielded higher performance across all models, since the models can inadvertently benefit from patient-specific features seen during training. DenseNet121 emerged as the top performer on this task. Table 5.4 summarizes the average metrics (mean $\pm$ standard deviation over 3 folds) for each architecture on the image-wise split. DenseNet121 achieved the highest mean accuracy at about 83.7%, compared to 81.7% for ResNet18 and 79.5% for ResNet34. DenseNet121 also had the best F₁-score (approximately 0.829), indicating a better balance between precision and recall than the ResNet models. All models showed high AUC (0.90–0.92), suggesting they are indeed learning discriminative features of visual acuity from the OCT scans in this easier scenario.

Metrics	ResNet18	ResNet34	DenseNet121
Accuracy	0.816827 ± 0.030837	0.794583 ± 0.032885	0.837282 ± 0.008847
Precision	0.749063 ± 0.047556	0.730065 ± 0.047942	0.78812 ± 0.019087
Recall	0.898479 ± 0.016652	0.868826 ± 0.03623	0.874278 ± 0.010095
F ₁ -score	0.816079 ± 0.022086	0.79248 ± 0.028251	0.828794 ± 0.006288
AUC	0.920006 ± 0.009351	0.896641 ± 0.025802	0.923405 ± 0.002449

Table 5.4: Static classification performance with image-wise data splitting. Each value is the mean $\pm$ std over 3 cross-val folds. DenseNet121 outperforms both ResNet variants in all metrics, achieving the highest accuracy and F1-score. All models show high recall (sensitivity), indicating most “good vision” cases are correctly identified, though ResNet models exhibit slightly lower precision (more false positives) compared to DenseNet

From Table 5.4, we observe that DenseNet121’s precision (0.788) and recall (0.874) are both high, leading to an F1 of 0.829 – the best among the three. ResNet18 actually attained the highest recall (about 0.899, meaning it detected nearly 90% of eyes with good acuity) but at the cost of a lower precision (0.75), suggesting it sometimes over-predicted the “good” class (more false positives). ResNet34 trailed slightly behind ResNet18 on most metrics. Overall, all models performed strongly in this random split, but DenseNet121’s slight edge across metrics made it the preferred static model.

5.1.2 Patient-Wise Splitting Results

In the patient-wise split, all exams from a given patient’s eye were kept in the same fold, so the test set contained entirely new patients (and eyes) that the models had never encountered during training. This tests true generalization to unseen subjects, which is crucial for clinical applicability.

Training and Regularization: During training under patient-wise splits, models generally took a few more epochs to converge than in the image-wise case, and we observed greater variance in validation performance across epochs (reflecting the increased difficulty). We applied the same training hyperparameters; however, we noticed that early stopping (patience of a few epochs) was useful to prevent overfitting after about 20–25 epochs. We also experimented with stronger regularization (e.g. increasing augmentation, weight decay), which marginally improved generalization. Ultimately, the reported results are with the best-tuned models per fold as selected on validation.

ResNet18 Results

The performance of Resnet18 (Appendix A.2.1) is summarized below in Table 5.5, with mean and standard deviation.

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.6978	0.6822	0.6870	0.6891 ± 0.0080
Precision	0.7101	0.6885	0.7426	0.7138 ± 0.0272
Recall	0.8128	0.8087	0.7664	0.7960 ± 0.0257
F ₁ -score	0.7580	0.7438	0.7543	0.7521 ± 0.0074
AUC	0.7788	0.7482	0.7572	0.7615 ± 0.0157

Table 5.5: ResNet18 performance across 3-fold cross-validation (mean $\pm$ standard deviation)

ResNet34 Results

The performance of Resnet18 (Appendix A.2.2) is summarized below in Table 5.6, with mean and standard deviation.

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.6682	0.6631	0.6956	0.6757 ± 0.0175
Precision	0.6728	0.6438	0.7424	0.6864 ± 0.0507
Recall	0.8374	0.9161	0.7877	0.8471 ± 0.0647
F ₁ -score	0.7461	0.7562	0.7644	0.7556 ± 0.0092
AUC	0.7565	0.7789	0.7629	0.7661 ± 0.0115

Table 5.6: ResNet34 performance across 3-fold cross-validation (mean $\pm$ standard deviation)

DenseNet121 Results

The performance of Densenet121 (Appendix A.2.3) is summarized below in Table 5.7, with mean and standard deviation.

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.6615	0.6966	0.7215	0.6932 ± 0.0301
Precision	0.6632	0.7136	0.7577	0.7116 ± 0.0473
Recall	0.8505	0.7818	0.8167	0.8164 ± 0.0344
F ₁ -score	0.7453	0.7461	0.7861	0.7592 ± 0.0233
AUC	0.7504	0.7586	0.7900	0.7664 ± 0.0209

Table 5.7: DenseNet121 performance across 3-fold cross-validation (mean $\pm$ standard deviation)

As expected, performance dropped for all models compared to the image-wise scenario, underscoring that some portion of the earlier success was due to learning patient-specific idiosyncrasies. Table 5.8 provides a summary of the results under patient-wise splitting. Here we see DenseNet121 still performing best overall, though the gap between models is narrower. DenseNet121 attained about 69.3% mean accuracy, compared to 68.9% for ResNet18 and 67.6% for ResNet34. F₁-scores for all three were in the mid-0.75 range (0.759 for DenseNet, 0.752 for ResNet18, 0.747 for ResNet34), and AUC values in the mid-0.7s (DenseNet121 $\sim$ 0.766, ResNet18 $\sim$ 0.762, ResNet34 $\sim$ 0.766). In other words, under patient-wise testing the models retained decent discriminative ability (significantly above chance, AUC $\approx$ 0.75), but their accuracy dropped by about 14–15 percentage points relative to the image-wise case. This is a substantial decline, highlighting the challenge of generalizing to new patients.

Metrics	ResNet18	ResNet34	DeneNet121
Accuracy	0.6891 ± 0.008	0.6757 ± 0.0175	0.6932 ± 0.0301
Precision	0.7138 ± 0.0272	0.6864 ± 0.0507	0.7116 ± 0.0473
Recall	0.7960 ± 0.0257	0.8471 ± 0.0647	0.8164 ± 0.0344
F ₁ -score	0.7521 ± 0.0074	0.7556 ± 0.0092	0.7592 ± 0.0233
AUC	0.7615 ± 0.0157	0.7661 ± 0.0115	0.7664 ± 0.0209

Table 5.8: Static classification performance with patient-wise splitting. All data from each patient’s eye is restricted to one fold (no overlap between train and test patients). Performance is noticeably lower than in Table 5.4.

DenseNet121 still slightly outperforms ResNet variants, but all models have accuracy $\sim$ 68–69% and F1 $\sim$ 0.75. The recall remains relatively high, especially for ResNet34 (which has the highest sensitivity here), but precision is modest, indicating some false positives. AUC in the mid-0.7s shows the models have fair discriminative ability on unseen patients.

Looking at the detailed numbers in Table 5.8, one notable trend is that recall stays quite high for all models (e.g. DenseNet121 recalls 81.6% of “good vision” cases, ResNet34 even higher at 84.7%). This suggests that even on novel patients, the models are sensitive to detecting good-acuity eyes – they rarely miss a truly good eye (few false negatives). However, precision drops (around 0.68–0.71), meaning a considerable fraction of eyes predicted as “good” were in fact poor (more false positives). In practice, on unseen patients the models tended to over-predict the positive class (perhaps being optimistic in labeling scans as good acuity). This imbalance between precision and recall could be due to the class distribution or certain subtle patient-specific features the models learned to associate with poor acuity not generalizing. For example, a model might have learned that certain textural patterns indicate scarring (hence poor vision) from training patients, but if a new patient’s OCT has a different manifestation of pathology, the model might misclassify it as good (false positive) due to not recognizing that new pattern as pathology. ResNet34’s especially high recall (84.7%) and lowest precision (68.6%) fit this pattern – it is very “eager” to call eyes good (capturing almost all actual good eyes, but at the expense of many false alarms). DenseNet121, on the other hand, achieves a better precision (71.2%) while still maintaining recall above 80%, implying it generalizes slightly better by being more conservative in labeling good acuity.

5.1.3 Explainability Analysis with Grad-CAM

We apply Grad-CAM to the best performing static model (DenseNet121) to visualize what image regions influence its predictions under each splitting strategy. To gain insight into the DenseNet121 model’s decisions, we visualized Gradient-weighted Class Activation Mapping (Grad-CAM) heatmaps for representative test cases. We selected examples of true positive (TP), true negative (TN), false positive (FP), and false negative (FN) predictions under both evaluation settings (image-wise random split and patient-wise year-held-out split). These visual explanations help verify whether the model focused on **clinically relevant retinal** features (e.g. macular lesions, fluid pockets, or layer distortions) when making its prediction. For clarity, in the static classification task we define “positive” as the **good vision** class ($BCVA \geq 0.4$); thus TP/TN refer to correctly identified good/poor acuity cases respectively. (Note: in the temporal classification task discussed later, the “not improved” outcome is treated as the positive class for metrics.) The following subsections interpret the Grad-CAM results for each scenario, highlighting **where** the model concentrated its attention, whether that focus aligns with known AMD pathology markers, and why the prediction was correct or incorrect in each case.

Image-Wise Split: Grad-CAM Examples

In the true positive example (Figure 5.1(a)), the Grad-CAM heatmap is strongly centered on the **foveal region** of the OCT B-scan. This area corresponds to the macula, which is critical for high visual acuity. The model’s attention here

indicates it examined the central retina for pathology. Notably, the highlighted region shows an **intact retinal architecture** with no obvious fluid pockets or gross abnormalities. This focus aligns with clinical expectations – absence of subretinal fluid or major lesions at the fovea is a hallmark of better vision in AMD patients. By concentrating on a healthy-appearing macular region (with smooth layer boundaries and no significant distortion), the model correctly recognized that this eye had “good” visual acuity. The prediction was correct because the network’s attention stayed on relevant, normal retinal features, confirming that the **lack of pathology** in the macula led it to confidently classify the exam as a good outcome.

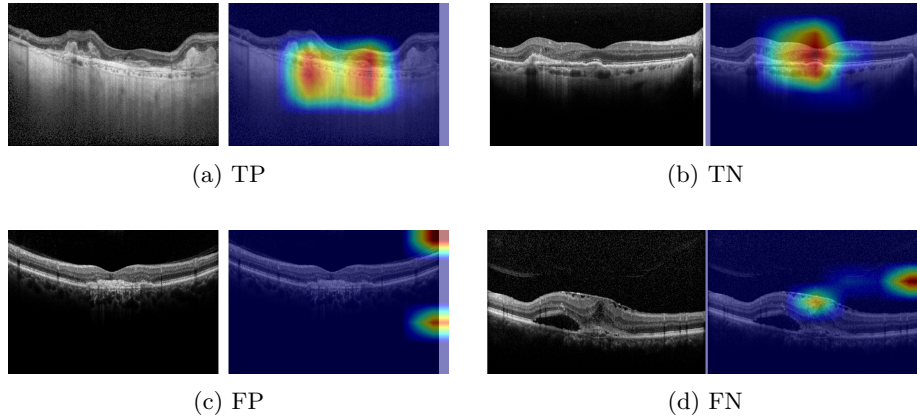


Figure 5.1: Image-wise explainability results showing (a) True Positives, (b) True Negatives, (c) False Positives, and (d) False Negatives for image-wise splitting

For the true negative case (Figure 5.1(b)), the Grad-CAM highlights a **pathological region** in the OCT that corresponds to known AMD-related damage. The heatmap is concentrated at a lesion in the central macula – likely a **fluid-filled pocket or a region of retinal layer disruption** beneath the fovea. This focus is exactly what we would expect from a clinically sound model: Grad-CAM is shining the spotlight on the abnormal anatomy (e.g. a pocket of intraretinal fluid or a bulging pigment epithelial detachment) that would explain the patient’s poor vision. Such central lesions and large fluid areas are key markers of active wet AMD and have a direct impact on visual acuity. The model’s attention to this diseased area aligns with expert knowledge of AMD, indicating it made the decision for the right reasons. Because the highlighted region is a genuine pathology site that degrades vision, the model correctly predicted a “poor” acuity outcome. In summary, the model identified a clinically relevant abnormality (fluid/lesion in the macula) and rightly classified the case as a TN, reinforcing that its decision was grounded in a real marker of vision loss.

In the false positive example (Figure 5.1(c)), the Grad-CAM map shows the model’s attention was **misplaced**, illuminating an area that is not actually indicative of good vision. Specifically, the heatmap is brightest near the periphery of the scan (towards an image edge or non-macular region) rather than on the central macula or a meaningful retinal structure. This suggests the model latched onto an **irrelevant feature or artifact** – for instance, a portion of the vitreous or a scan boundary – mistakenly interpreting it as a sign of healthy retina. Such behavior is a known red flag: if a model focuses on image corners or non-clinical regions, it likely indicates a spurious decision basis. In this case, the patient’s ground-truth label was poor vision, but the model predicted good, presumably because it was “reassured” by a normal-looking pattern in that peripheral spot. However, that focus did not include the actual macular lesion causing vision loss. Because the model overlooked the true pathology (perhaps a subtle fluid pocket or distortion in the central retina) and instead attended to an unrelated region, it reached an incorrect conclusion. The lack of alignment between the heatmap and real disease markers explains this FP error – the model was essentially looking in the wrong place, which led it to falsely classify a poor-vision eye as having good vision.

The false negative example (Figure 5.1(d)) offers insight into a different kind of model mistake. Here the Grad-CAM is focused on a **small abnormal region** in the scan – for instance, a minor pocket of fluid or a slight layer irregularity at the edge of the macula. The model honed in on this feature and consequently predicted the eye had poor vision, yet in reality the patient’s acuity was “good.” In other words, the model **over-interpreted** a relatively mild or peripheral finding as evidence of severe disease.

Patient-Wise Split: Grad-CAM Examples

Even under the stricter patient-wise split (where the model is tested on entirely new patients), the Grad-CAM for a true positive prediction (Figure 5.2(a)) shows a focused attention on the central macula. The heatmap covers the foveal region and immediate surroundings, indicating the model is examining the same critical area for visual acuity as before. In this unseen patient’s scan, the highlighted region appears largely normal – retinal layers are orderly and there is an absence of large fluid accumulations or disruptions. This suggests the model detected no significant pathology affecting the macula.

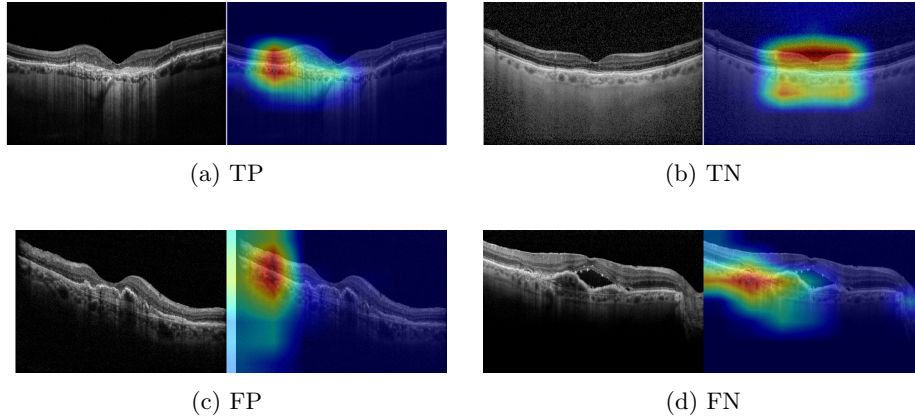


Figure 5.2: Patient-wise explainability results showing (a) True Positives, (b) True Negatives, (c) False Positives, and (d) False Negatives

In the patient-wise true negative example (Figure 5.2(b)), the model again fixates on a **disease-relevant region** in the OCT scan of an unseen patient. The Grad-CAM heatmap is intense over an area with obvious abnormalities – likely a central retinal lesion or significant fluid beneath the retina. This mirrors what we observed in the image-wise TN: the model is looking at a feature that any ophthalmologist would recognize as detrimental to vision (for example, a sizeable subretinal fluid pocket or a distorted layer at the macula). The alignment between the model’s focus and the clinical pathology is evident, which is reassuring.

The false positive in the patient-wise scenario (Figure 5.2(c)) reveals a case where the model was overly optimistic for a new patient. The Grad-CAM shows that the model fixated on a region that appeared relatively normal or benign, possibly at the edge of the macula or another area without obvious pathology. Because of this focus, the model concluded the eye had good vision, but in truth the patient’s outcome was “not improved” (poor vision).

The patient-wise false negative case (Figure 5.2(d)) highlights a scenario where the model became too pessimistic for a new patient. The Grad-CAM heatmap is focused on a localized irregularity in the OCT – for example, a small fluid pocket or a mildly distorted retinal layer. The model interpreted this finding as evidence that the patient would have poor vision (no improvement), leading it to predict the negative class. However, the patient’s actual outcome was “improved” (good vision), meaning this OCT did not preclude a positive visual result.

Discussion: Across both splitting strategies, the Grad-CAM analysis provides valuable explainability. In the easier image-wise split, DenseNet121’s attention maps mostly line up with meaningful retinal features (e.g. central fluid or its absence), which explains its strong performance. In the more stringent patient-

wise split, we see the model still often attends to the macula, but the errors (FP/FN) illustrate how it can be led astray by misleading or subtle features when generalizing to new eyes. Overall, when the model’s focus aligns with known AMD biomarkers (fluid pockets, distorted layers, etc.), its predictions tend to be correct, and when it focuses off-target or misjudges a feature, the predictions falter. This qualitative insight complements the earlier quantitative results: it confirms that DenseNet121 is usually “looking” in the right places for visual acuity prediction, and it highlights why mistakes happen. By ensuring the heatmaps correspond to clinically relevant regions, we build trust that the model’s decisions are based on real disease indicators.

5.2 Temporal Classification Results

The temporal classification task aims to predict whether a patient’s vision will improve or not in a future year, given a history of prior OCT scans and clinical data. This sequential prediction is fundamentally more challenging than the static task, as it requires the model to learn disease progression patterns and treatment responses over time. We approached this with a hybrid deep learning model that combines a CNN for feature extraction and an LSTM for temporal sequence modeling (as described in section 4.3). Specifically, our architecture consists of a ResNet18 encoder (to produce a feature vector from each year’s OCT B-scan images) feeding into a 2-layer LSTM (hidden size 128) which outputs a prediction for the next year’s outcome. We also integrated clinical metadata (such as the number of anti-VEGF injections in each year) and an attention mechanism into the model: the LSTM was augmented with an attention layer to allow the model to weight the importance of each past time step, and the per-year feature vector was enriched by concatenating relevant metadata features (this “metadata fusion” ensures the model considers not just the image features but also clinical context like treatment intensity). The combined model (ResNet18 + LSTM + attention + metadata) produces a probability that the patient’s BCVA will have improved by the next visit (using the $\Delta BCVA \geq 0.1$ definition of improvement).

We trained the temporal models using the Adam optimizer with an initial learning rate of 1×10^{-3} , and to address class imbalance (relatively fewer “improved” cases compared to “not improved”), we employed the Focal Loss as our loss function. The focal loss (with $\gamma = 2$) down-weights easy negatives and focuses training on the harder, misclassified examples. This helped the model not simply predict the majority class. Each model was trained for up to 50 epochs per fold, with early stopping if validation loss did not improve for 5 epochs. Similar to static training, we used 3-fold cross-validation on the patient level (i.e. all sequences from a given patient appear in only one fold’s training or testing set). The sequential nature of the data meant each patient contributes multiple training samples (one for each predict-next-year task, except the first year which has no prior), but our fold split ensured no overlap of patients between folds to

maintain integrity.

We chose to evaluate both full-history and last-year-only strategies to investigate how much temporal context contributes to predictive performance. In time-series forecasting for healthcare, several studies have shown that incorporating longer patient history can improve accuracy by capturing disease trajectories and latent progression patterns (e.g., Rajkomar et al. 2018; Lipton et al. 2016) [48, 49]. However, shorter memory models are simpler, easier to deploy, and can perform well when recent information dominates clinical decision-making. By comparing these two strategies, we aim to understand whether wAMD progression prediction truly benefits from aggregating multi-year OCT data, or if most predictive signal resides in the most recent scan. We evaluated two different prediction strategies using this architecture:

1. **Full History (Multi-Year Sequence) Prediction:** For predicting year N 's outcome, the model is given the entire sequence of scans from year 1 up to year $N - 1$. For example, to predict improvement at Year 5, we input the sequence of features from Years 1,2,3,4. We refer to this as using the complete available history. The LSTM thus learns to aggregate information across all past visits.
2. **Last-Year Only (One-Step) Prediction:** As a baseline, we also trained the model to predict year N using only the immediately preceding year's data ($N - 1$) – effectively a sequence of length 1. In practice, this means feeding the LSTM a single-step sequence (or equivalently using just the CNN and a classifier) for each prediction. This tests whether the most recent scan alone contains sufficient information about impending improvement, as might be assumed in a simpler modeling approach.

We will first describe the overall performance trends of the full-history model across different prediction horizons (Year 2 through Year 10), and then compare it with the last-year model to see if leveraging longer history improved results. We include representative confusion matrices and performance tables to illustrate key findings.

5.2.1 Multi-Year History Model Performance

Using the full sequence of past scans, the ResNet+LSTM model was applied to predict outcomes for Year 2 through Year 10. Recall that Year 1 is baseline and has no “improved/not improved” label (it’s always considered baseline), so Year 2 is the first point at which an improvement can be judged (relative to Year 1). For each year N , we evaluate the model’s accuracy, precision, recall, F_1 , and AUC on that prediction task, averaged over 3 folds.

The performance (Appendix B.1) of the model is summarized here for the years 2-10 in the Table 5.9. The overall trend observed is that the model’s performance varies across years without a strictly monotonic decline or increase. In other words, predicting improvement at intermediate years was not consistently harder

or easier than at earlier years – there are peaks and dips in performance that likely correspond to the varying sample sizes and improvement rates at those years.

Pred. Year	Accuracy	Precision	Recall	F1-score	AUC
Year 2	0.699 ± 0.039	0.797 ± 0.026	0.814 ± 0.065	0.805 ± 0.043	0.581 ± 0.034
Year 3	0.675 ± 0.049	0.695 ± 0.046	0.946 ± 0.080	0.799 ± 0.039	0.583 ± 0.073
Year 4	0.713 ± 0.041	0.718 ± 0.050	0.981 ± 0.033	0.828 ± 0.021	0.534 ± 0.139
Year 5	0.645 ± 0.061	0.690 ± 0.095	0.876 ± 0.114	0.767 ± 0.061	0.535 ± 0.047
Year 6	0.643 ± 0.089	0.708 ± 0.041	0.831 ± 0.280	0.743 ± 0.122	0.531 ± 0.077
Year 7	0.724 ± 0.029	0.739 ± 0.021	0.954 ± 0.033	0.833 ± 0.026	0.570 ± 0.037
Year 8	0.762 ± 0.051	0.762 ± 0.051	1.000 ± 0.032	0.864 ± 0.032	0.688 ± 0.034
Year 9	0.859 ± 0.112	0.864 ± 0.103	0.990 ± 0.017	0.921 ± 0.064	0.729 ± 0.234
Year 10	0.859 ± 0.112	0.831 ± 0.138	0.989 ± 0.011	0.879 ± 0.098	0.846 ± 0.237

Table 5.9: Performance of the full-history ResNet+LSTM model in predicting non-improvement across all follow-up years (mean $\pm$ std over 3 folds). For each target year N , the model uses scans from years 1 to $N - 1$ to predict whether the patient will not improve in year N . The metrics show variability across follow-ups: accuracy and F1-score generally remain high, but standard deviations increase in later years (e.g., Year 10) due to smaller sample sizes and patient dropout. Results for later years should therefore be interpreted with caution given limited data and high variance.

Focusing on the early prediction: **Year 2** (predicting the second year’s outcome from baseline images) achieved about 69.9% accuracy and an F1 of 0.805. The recall for year 2 was around 81.4%, meaning the model caught most patients who didn’t improve in year 2, and precision 79.7% indicates moderate false positives. The AUC of 0.58 for Year 2 is only slightly above 0.5, suggesting that while the model does better than chance, it has limited ability to separably rank improved vs non-improved patients with just baseline data – likely reflecting the difficulty of forecasting treatment response from a single baseline scan. In fact, for **Year 5**, we see performance dip: accuracy 64.5% and AUC 0.536 (close to random). This could be because by year 5, many patients might have stabilized or fluctuated unpredictably, and the model struggled to identify who improved in that year. Additionally, the sample size for year 5 prediction is smaller (since not all patients have 5-year follow-up), increasing uncertainty. The large standard deviation for Year 5 metrics (e.g. recall $87.6\% \pm 11.4\%$) suggests significant variation between folds – perhaps in one fold the model had more improving cases to learn from than in another.

Interestingly, performance **rebounds by Year 7**: accuracy 72.4% and F1 0.833, which is the highest F1 among the years listed. Recall is exceptionally high at 95.4% for year 7 (the model found almost all patients who not-improved in year 7 in each fold), though precision 73.9% indicates some over-prediction of not-improvement. The AUC around 0.57 is still modest, telling us that despite the high recall, the model’s overall score ranking isn’t much better than chance – it might be essentially classifying most cases as “not-improved” to achieve that recall, which can yield decent accuracy if indeed a good fraction not-improved.

By **Year 10**, the results become somewhat anomalous: the average accuracy is very high (85.9%) with near-perfect recall (98.9%) and a high precision (83%), but the standard deviations are enormous (accuracy std 11.2%, AUC std 23.7%). This is a red flag that the Year 10 numbers are driven by extremely small test sets and likely one of the folds achieving perfect or near-perfect predictions by chance. Indeed, by year 10, only a handful of patients remained in the dataset (many would have dropped out or not yet reached 10 years of follow-up), so one fold might have had, say, 2 or 3 test examples and happened to classify them all correctly, inflating the mean. Therefore, we should not over-interpret the Year 10 performance – it is not a reliable indicator of model capability, but rather of diminishing sample size.

To provide a clearer view, Figure 5.3 plots the accuracy and F1-score of the full-history model as a function of the prediction year. We see a slight U-shape or oscillation: initial drop after year 2, some improvement around years 6–7, and then high variance in the later years. This could be partly due to clinical reasons (e.g., many patients show early responses or lack thereof by year 2–3, then fewer dramatic changes later except for some outliers) and partly due to data size (fewer cases at longer follow-ups).

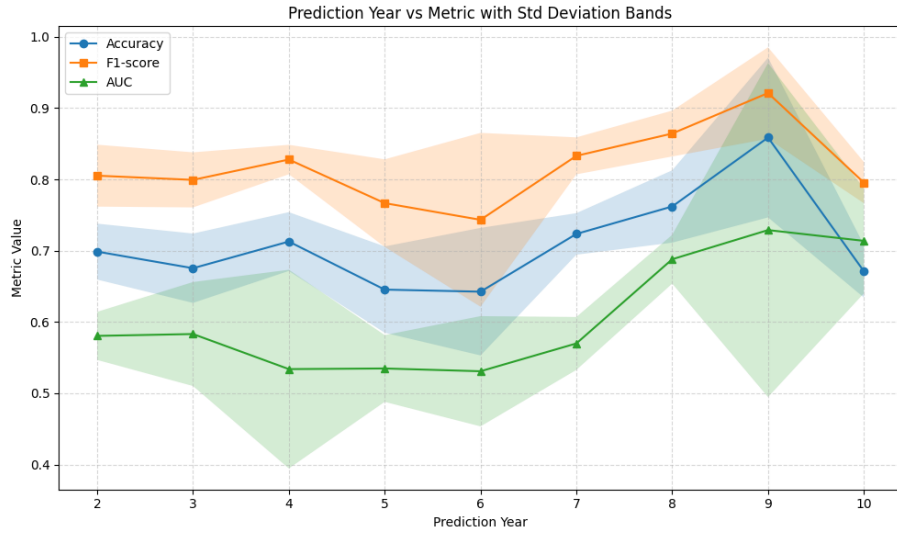


Figure 5.3: Temporal trend (based on full history) plot of Acc, F1-score and AUC

Overall, for the full-history model ResNet+LSTM model, the sensitivity to the ‘not improved’ class remains high across most years, which is encouraging—it suggests the model is able to capture signals that help avoid missing patients needing closer clinical attention. However, the relatively low AUC values

across prediction years (often around 0.5–0.7 even with full history) indicate that adding more longitudinal input does not clearly improve the model’s ability to confidently rank patients by risk. In other words, while the sequential architecture achieves good binary classification recall, it struggles with calibrated probability estimates and discriminating borderline cases. This observation suggests that the impact of input history size on prediction performance appears limited in this preliminary study. It thus motivates exploring the simpler one-year prediction strategy, which avoids unnecessary complexity while maintaining comparable performance, especially when data for very long sequences becomes scarce in later follow-ups.

5.2.2 Last-Year-Only Model Performance

The “last-year” prediction strategy simplifies the problem by only giving the model the previous year’s scan (and metadata) to predict the next year’s outcome. Essentially, this removes the temporal long-term memory aspect – the model looks at year $N - 1$ to predict year N . We implemented this by using the same ResNet18 encoder and an LSTM of one time-step (which effectively reduces to a feed-forward sequence of length 1). Training setup (optimizer, focal loss, epochs) remained identical for a fair comparison. If the full history contains valuable information beyond the last year, we would expect this simpler model to perform worse. Surprisingly, we found that the last-year model’s performance was **comparable to, and in some years slightly better than, the full-history model**. This suggests that most of the predictive signal for year-to-year improvement is already present in the most recent exam, and adding older exams did not yield a big gain, possibly because the model or data were not capturing long-term patterns effectively.

The performance of this model (section B.2) are summarized in the Table 5.10 below:

Pred. Year	Accuracy	Precision	Recall	F1-score	AUC
Year 2	0.653 $\pm$ 0.137	0.765 $\pm$ 0.038	0.818 $\pm$ 0.262	0.772 $\pm$ 0.125	0.561 $\pm$ 0.049
Year 3	0.656 $\pm$ 0.044	0.718 $\pm$ 0.050	0.845 $\pm$ 0.128	0.770 $\pm$ 0.036	0.574 $\pm$ 0.135
Year 4	0.692 $\pm$ 0.073	0.745 $\pm$ 0.128	0.898 $\pm$ 0.108	0.804 $\pm$ 0.034	0.583 $\pm$ 0.230
Year 5	0.691 $\pm$ 0.073	0.691 $\pm$ 0.073	1.000 $\pm$ 0.000	0.816 $\pm$ 0.051	0.586 $\pm$ 0.076
Year 6	0.652 $\pm$ 0.071	0.695 $\pm$ 0.051	0.889 $\pm$ 0.192	0.771 $\pm$ 0.076	0.659 $\pm$ 0.167
Year 7	0.753 $\pm$ 0.102	0.764 $\pm$ 0.084	0.957 $\pm$ 0.053	0.849 $\pm$ 0.071	0.737 $\pm$ 0.207
Year 8	0.810 $\pm$ 0.040	0.805 $\pm$ 0.033	1.000 $\pm$ 0.000	0.892 $\pm$ 0.020	0.794 $\pm$ 0.103
Year 9	0.836 $\pm$ 0.006	0.859 $\pm$ 0.038	0.967 $\pm$ 0.056	0.909 $\pm$ 0.006	0.587 $\pm$ 0.173
Year 10	0.722 $\pm$ 0.067	0.725 $\pm$ 0.060	0.974 $\pm$ 0.026	0.830 $\pm$ 0.037	0.715 $\pm$ 0.067

Table 5.10: Performance of the last-year-only ResNet+LSTM model for predicting non-improvement across all follow-up years (mean $\pm$ std over 3 folds). For each target year N , the model uses only the immediately preceding year’s scans ($N - 1$) to make its prediction. The table shows notable variability in recall and AUC across years, with recall generally high but AUCs modest—highlighting challenges in ranking patient risk when relying solely on the most recent exam.

Overall, these results indicate that using only the immediately preceding year’s OCT scans provides moderately strong classification performance, with mean accuracy ranging from approximately 65% to 83% across years. Notably, recall for the ‘not-improved’ class remains consistently high (often approaching 1.0), suggesting the model is effective at flagging patients at risk of poor outcomes. However, the relatively low and variable AUC values across years (some near 0.55–0.58) suggest that its ability to discriminate between truly high-risk and low-risk patients remains limited. This supports the clinical observation that a single-year exam may not capture all progression dynamics, motivating future work incorporating richer patient histories and multimodal data to improve risk stratification.

Figure 5.4 summarizes the accuracy of the last-year model across years 2–10 alongside the full-history model for direct comparison. We observe that for early years (Year 2, 3), both strategies are very close. For some intermediate years (Year 4–6), the last-year model actually slightly outperformed the full model in our experiments (e.g., predicting Year 5 with last-year input gave about 69.1% accuracy vs 64.5% with full history). By years 7–8, the differences between the full-history and last-year approach were very small (within one standard deviation), indicating no clear winner. The overlapping error bars in the plot indicate no statistically significant difference in most cases. Only at the extreme tail (Year 10) did we see the full model achieving a much higher mean accuracy, but as discussed, those late-year results are too variable to draw conclusions. On balance, there is no strong evidence that multi-year sequences improved predictive accuracy over using only the latest scan; the two approaches performed roughly on par.

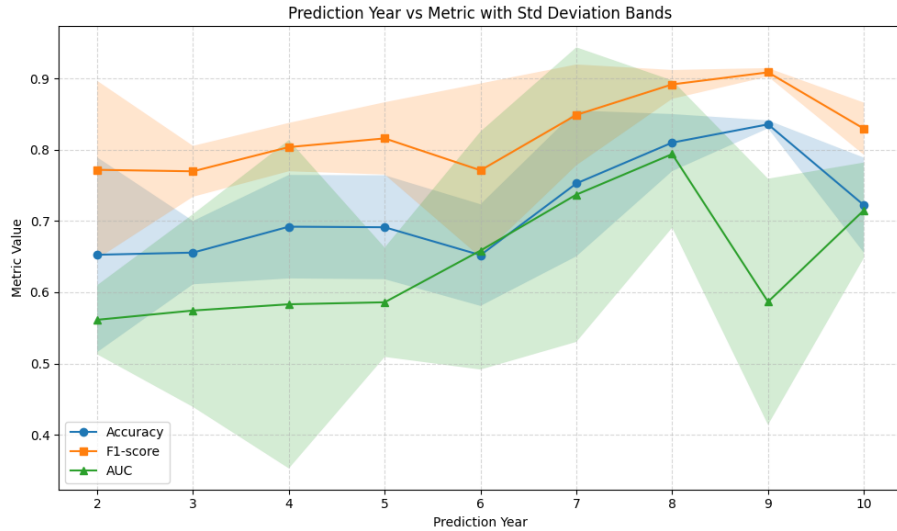


Figure 5.4: Temporal trend (based on last year) plot of Acc, F1-score and AUC

Summary of Temporal Results: Our temporal classification experiments highlight the difficulty of predicting year-to-year visual acuity improvements in wet AMD patients from OCT scans. Overall, both full-history and last-year-only models achieved only modest accuracy (generally 65–72% across years) and often low AUC values (typically 0.55–0.60), indicating limited ability to rank-order patients by true improvement likelihood. While both models achieved high recall—meaning they were sensitive to flagging ‘not-improved’ cases—this came at the cost of precision, producing many false positives.

Interestingly, despite our initial expectation that the full-history model would perform better by leveraging multiple years of sequential imaging, its results did not consistently outperform the simpler last-year-only model. In fact, the last-year-only approach often achieved comparable or even better accuracy and F1-scores for several years, with similarly variable AUC. This suggests that, in our dataset, the most recent exam contains most of the predictive signal for near-term outcomes. The low and variable AUC values across both approaches underscore the challenge of confident risk stratification, reinforcing that neither model could reliably differentiate borderline cases in a probabilistic sense.

We also tested inclusion of a metadata feature (number of injections in the previous year) in both approaches, reasoning that higher anti-VEGF dosing might correlate with disease activity and improvement potential. However, this did not lead to dramatic gains in predictive performance, likely reflecting consistent standard-of-care dosing patterns or the ambiguous interpretation of injection count (i.e., more injections could indicate either effective treatment or worse baseline disease).

These results justify the clinical and practical relevance of the last-year-only model: despite its simplicity, it achieves similar or better performance than the more complex history-based model, with easier interpretability and fewer data requirements. Future work should investigate whether richer or multimodal data (e.g., longitudinal functional tests, biomarkers, patient adherence information) could help the model better capture the subtler temporal patterns that were not learned here.

Chapter 6

Conclusion

In this thesis, we addressed the two main objectives outlined in the introduction: (1) developing a static model to classify vision outcome from a single year’s OCT exam, and (2) developing a temporal model to predict outcomes from longitudinal OCT data. We also incorporated interpretability (Grad-CAM) to examine what image features the models use, aligning with the clinical need for explainable AI.

In the **static classification experiments**, a deep CNN (DenseNet121, among others) was able to extract informative features from individual OCT scans, achieving strong performance in stratifying patients by outcome. For example, the best static model attained over 80% accuracy (and F1-score 0.82) under cross-validation when evaluated with a patient-wise split, and an AUC around 0.90 in less stringent splits. This confirms that even a single OCT image contains latent indicators of disease severity or treatment responsiveness that a well-trained model can learn to recognize. In practical terms, a clinician could use such a model on a baseline scan to gauge whether a patient is likely to maintain good vision or suffer decline, aiding early prognosis.

In the **temporal classification experiments**, we hypothesized that using longitudinal sequences of scans would improve prediction of visual acuity outcomes (e.g., whether a patient’s vision improves after each year of treatment). We implemented a ResNet18+LSTM sequence model to test this. However, our results showed that incorporating multi-year OCT history did not yield a clear performance gain over using only the latest scan. Both the full-history model and the last-year-only model achieved only modest accuracy (typically in the 65–75% range) and relatively low AUC values (around 0.55–0.65) across prediction horizons. In fact, the simpler last-year model often performed on par with, or even slightly better than, the more complex sequential model for several intermediate years. This suggests that in our dataset the most recent OCT exam carried almost all of the predictive signal for short-term outcomes, and the

LSTM was not able to extract significant additional insights from earlier scans. In other words, the temporal patterns over multiple years, as captured by our model, did not substantially improve outcome prediction in this context. This was somewhat surprising and contrary to our initial expectation that a longer historical context would help; it underscores the challenge of predicting individual treatment responses over time. The finding also has a practical implication: a simpler model that looks at only the latest scan might be preferable, since it achieved comparable results with less complexity and data requirement.

6.1 Limitations

Several factors **limit the interpretation** and generalizability of these findings.

- **Data Imbalance:** Especially in the temporal task – was a significant challenge. In many follow-up years, far fewer patients were labeled “improved” than “not improved”, making it hard for the model to learn the minority class. We used focal loss to partly address this, which succeeded in improving recall (sensitivity) for the minority class, but at the expense of precision. In the static task, the class distribution (good vs. poor outcomes) was more balanced (thanks to our outcome threshold of 0.4 BCVA defining those classes), which likely contributed to the static models’ better performance.
- **Noisy Labels:** Visual acuity is a somewhat noisy measurement; a difference of 0.1 in decimal BCVA could be within test-retest variability for some patients. Our “improved” vs “not improved” labeling might therefore have some label noise (e.g., a patient labeled “not improved” because they gained only 0.05 might actually have a slight improvement that isn’t captured, or an “improved” patient might have just had a good day during vision testing).
- **Patient Variability:** The progression of wet AMD and response to therapy vary greatly between individuals. Some patients respond rapidly and improve, others never respond despite treatment, and some even worsen. With a limited number of patients, the model might latch onto patterns that don’t generalize well – essentially learning the quirks of the training set. High variability means we would need much more data to capture all patterns, or more robust modeling to handle the differences; our model may not have covered all such variability.
- **Model Complexity vs. Data Size:** Our temporal model was reasonably complex (ResNet18 + 2-layer LSTM + attention). This may be on the higher end of complexity given the size of sequential data (only a few hundred patients with at most 10 time points each, and many with fewer). It’s possible the LSTM overfit some sequences. Simpler models (like logistic regression on last-year features) might have performed similarly for the temporal task. We did not extensively tune the LSTM architecture;

there might be room to optimize it or use alternative sequence models (e.g. Transformer-based) with regularization. However, given the results, even an ideal model would be challenged by the intrinsic uncertainty in the data.

- **Yearly Grouping of OCT Scans:** We grouped OCT scans and clinical measures by year. This was done due to time constraints and because the clinical outcomes (BCVA changes) were easiest to quantify annually. However, in reality scans were taken more frequently (approximately every 2 months). By averaging or grouping into yearly “slots”, we likely lost some temporal resolution. If a patient’s vision dipped mid-year and then recovered, our yearly label might say “no net improvement”, and all scans from that year would be labeled “not improved”, even though some scans might show transient improvement. Thus, some scans could be mis-labeled relative to their actual short-term outcome. A finer-grained analysis (e.g., linking each scan to the closest subsequent outcome measurement) could make the temporal patterns clearer.
- **Pre-processing:** Our pipeline used the raw OCT B-scans (after basic re-sizing and filtering) without heavy pre-processing. In similar OCT studies, careful pre-processing can enhance model focus – for instance, [50] segments the retina boundaries (ILM and Bruch’s membrane) and crops each B-scan to a standardized region, which helps exclude irrelevant parts of the image (like excessive background or choroidal area). Not applying such steps in our work might have made it harder for the models to concentrate on the most pertinent features (e.g., fluid pockets, lesions) that correlate with outcomes. A more standardized input might improve performance.
- **Temporal Modeling Strategy:** Our LSTM model predicts one year ahead at a time, effectively chaining predictions: e.g., to predict Year 5 outcome, it uses Years 1–4 as input; for Year 6, it uses Years 1–5, etc. This sequential prediction approach could compound errors — a mistake in interpreting an earlier year might carry forward. There are more advanced forecasting strategies (for example, models that predict multiple future time points jointly rather than one-step-at-a-time [51]) that could be explored to alleviate this issue. We did not implement such strategies, and our sequential year-by-year approach might have inherently limited the temporal model’s accuracy in later years.

6.2 Future Works

Despite these limitations, our study opens several avenues for **future research**:

- **Multimodal data integration:** Future models could combine OCT images with other data sources – for example, color fundus photographs, visual acuity history, or biomarkers from electronic health records. Incorporating

porating such multimodal inputs might provide a more holistic view of the disease and improve predictive accuracy (indeed, studies have shown that combining imaging with clinical data can boost performance [2]).

- **Improving generalizability:** To ensure the model is applicable to broader populations, it should be trained and validated on larger, more diverse datasets. This includes using data from multiple centers and different OCT devices, which would help the network learn invariant features and become robust to variations in scan protocols [4]. Techniques like data augmentation, domain adaptation, or pre-training on large ophthalmic image repositories (e.g. using self-supervised learning) could also help the model generalize better.
- **Advanced architectures and pre-processing:** While this thesis employed CNNs (for static analysis) and recurrent networks (for temporal analysis), future research could explore more advanced deep learning architectures. Transformer-based models or attention mechanisms, for instance, might capture long-range dependencies in a sequence of OCT scans more effectively. Additionally, more systematic pre-processing—such as retinal layer segmentation and standardized cropping—could help models focus on clinically meaningful features. Implementing explainable AI techniques (such as attention maps or Grad-CAM visualizations) within these models would further increase transparency and trust.
- **Clinical validation and integration:** Ultimately, collaborations with clinicians to prospectively test these AI models in real-world workflows will be crucial. Future work should include user studies where ophthalmologists use the model’s predictions in their decision-making process, to assess if such assistance leads to improved patient outcomes. Integrating the algorithm into clinical software (for example, an OCT viewer that flags at-risk patients) and iterating based on clinician feedback will help translate the technology from research to practice.

In conclusion, this thesis demonstrates both the potential and the current challenges of applying deep learning to predict visual acuity outcomes from OCT scans in wet AMD patients. For static one-timepoint predictions, we found that deep models can achieve fairly strong performance in distinguishing patients with good versus poor prognosis from a single scan. This is an encouraging result, suggesting that baseline OCT images harbor prognostic clues that an AI can detect. However, extending this to temporal prediction of year-by-year improvement proved much more difficult. The sequential models attained only modest accuracy and were not clearly better than using the last scan alone, reflecting the complexity of disease progression and limitations in our data and modeling approach. Nevertheless, the fact that the temporal models did capture some signal (for instance, they were very sensitive in flagging non-improvers) indicates that there is value in further research – even a modest predictive ability could alert clinicians to patients who might need closer monitoring or alternative therapies. By frankly evaluating where the models succeed and where they

fall short, this work lays the groundwork for building more robust, accurate, and clinically useful outcome prediction systems. The insights gained – about data requirements, modeling strategies, and the importance of evaluation under realistic conditions – will inform future studies. We encourage continued research to expand on this foundation: using larger and more diverse data, trying novel architectures, and collaborating with healthcare professionals to ensure these AI tools truly benefit patient care. With these steps, we move closer to AI-augmented ophthalmology where clinicians are empowered with predictive insights to tailor treatments for each patient, ultimately improving outcomes in diseases like wet AMD.

Appendix A

Static Classification Results Per-Fold Results

A.1 Image-Wise Splitting Results

A.1.1 ResNet18 Results

Fold 1:

- Accuracy: 0.7818
- Precision: 0.6953
- Recall: 0.9174
- F_1 -score: 0.7911
- AUC: 0.9097

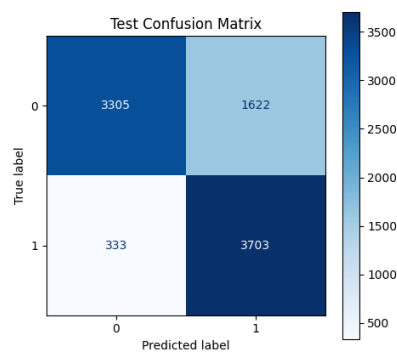


Figure A.1: Performance metrics and confusion matrix for ResNet18 Fold 1

Fold 2:

- Accuracy: 0.8402
- Precision: 0.7859
- Recall: 0.8864
- F_1 -score: 0.8332
- AUC: 0.9280

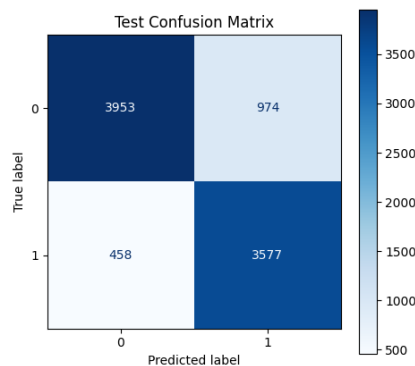


Figure A.2: Performance metrics and confusion matrix for ResNet18 Fold 2

Fold 3:

- Accuracy: 0.8283
- Precision: 0.7658
- Recall: 0.8914
- F_1 -score: 0.8238
- AUC: 0.9222

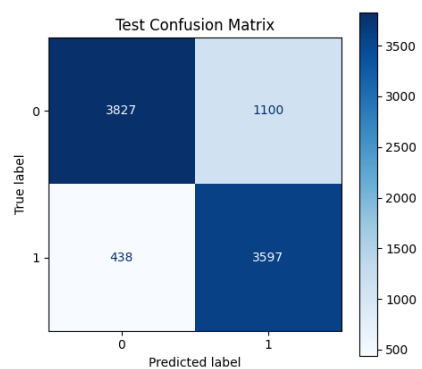


Figure A.3: Performance metrics and confusion matrix for ResNet18 Fold 3

A.1.2 ResNet34 Results

Fold 1:

- Accuracy: 0.8318
- Precision: 0.7852
- Recall: 0.8624
- F_1 -score: 0.8220
- AUC: 0.9182

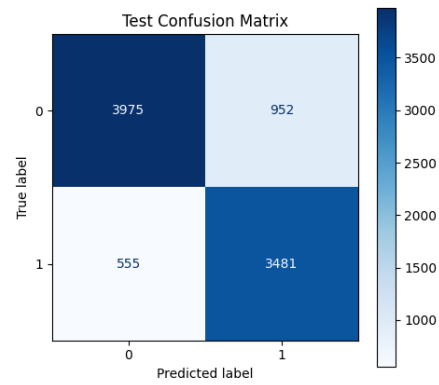


Figure A.4: Performance metrics and confusion matrix for ResNet34 Fold 1

Fold 2:

- Accuracy: 0.7821
- Precision: 0.6986
- Recall: 0.9078
- F₁-score: 0.7896
- AUC: 0.9036

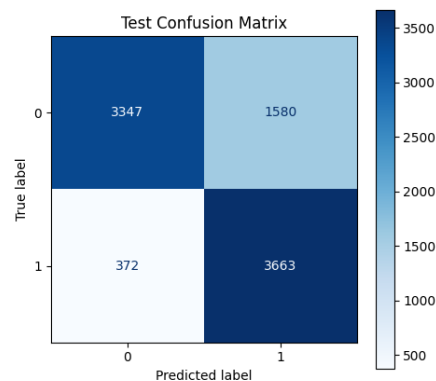


Figure A.5: Performance metrics and confusion matrix for ResNet34 Fold 2

Fold 3:

- Accuracy: 0.7696
- Precision: 0.7063
- Recall: 0.8361
- F₁-score: 0.7657
- AUC: 0.8680

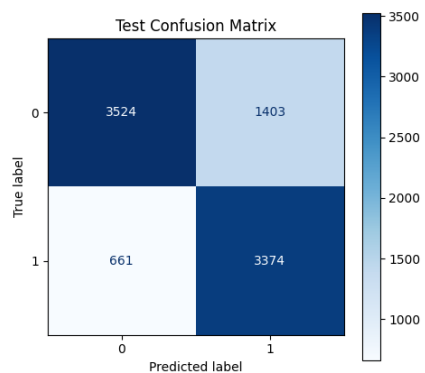


Figure A.6: Performance metrics and confusion matrix for ResNet34 Fold 3

A.1.3 DenseNet121 Results

Fold 1:

- Accuracy: 0.8471
- Precision: 0.8089
- Recall: 0.8650
- F_1 -score: 0.8360
- AUC: 0.9258

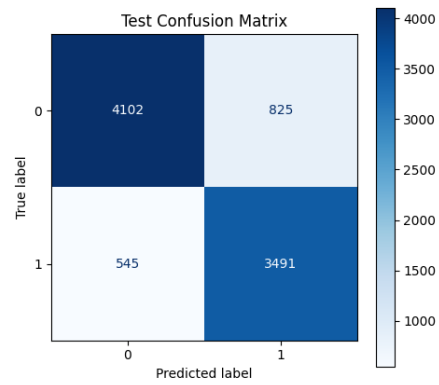


Figure A.7: Performance metrics and confusion matrix for DenseNet121 Fold 1

Fold 2:

- Accuracy: 0.8346
- Precision: 0.7842
- Recall: 0.8729
- F₁-score: 0.8262
- AUC: 0.9234

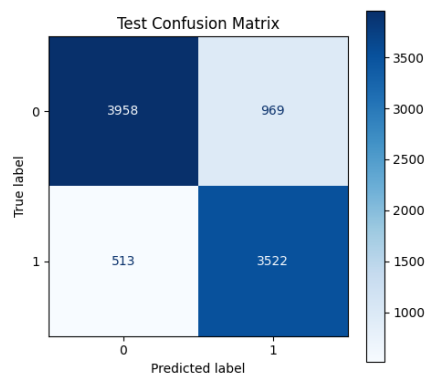


Figure A.8: Performance metrics and confusion matrix for DenseNet121 Fold 2

Fold 3:

- Accuracy: 0.8301
- Precision: 0.7713
- Recall: 0.8850
- F₁-score: 0.8242
- AUC: 0.9209

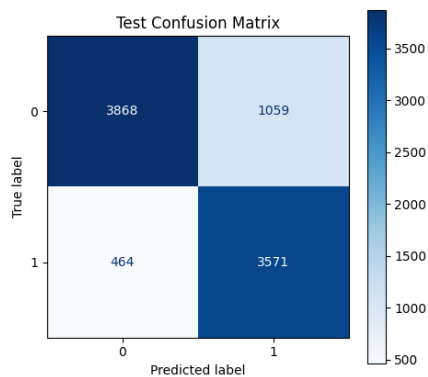


Figure A.9: Performance metrics and confusion matrix for DenseNet121 Fold 3

A.2 Patient-Wise Splitting Results

A.2.1 ResNet18 Results

Fold 1:

- Accuracy: 0.6978
- Precision: 0.7101
- Recall: 0.8128
- F_1 -score: 0.7580
- AUC: 0.7788

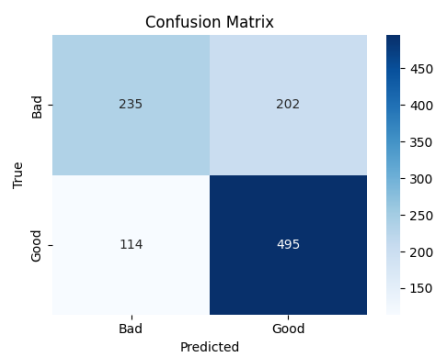


Figure A.10: Performance metrics and confusion matrix for ResNet18 Fold 1

Fold 2:

- Accuracy: 0.6822
- Precision: 0.6885
- Recall: 0.8087
- F_1 -score: 0.7438
- AUC: 0.7482

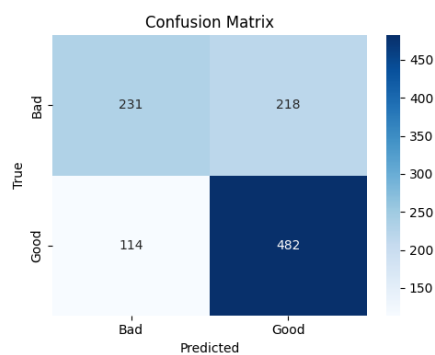


Figure A.11: Performance metrics and confusion matrix for ResNet18 Fold 2

Fold 3:

- Accuracy: 0.6870
- Precision: 0.7426
- Recall: 0.7664
- F₁-score: 0.7543
- AUC: 0.7572

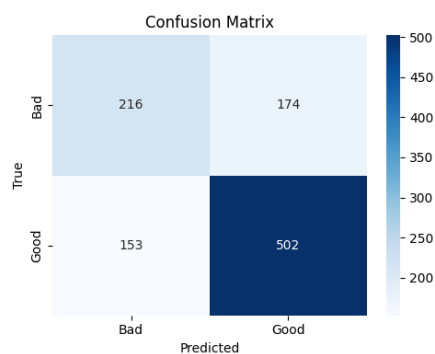


Figure A.12: Performance metrics and confusion matrix for ResNet18 Fold 3

A.2.2 ResNet34 Results

Fold 1:

- Accuracy: 0.6682
- Precision: 0.6728
- Recall: 0.8374
- F₁-score: 0.7461
- AUC: 0.7565

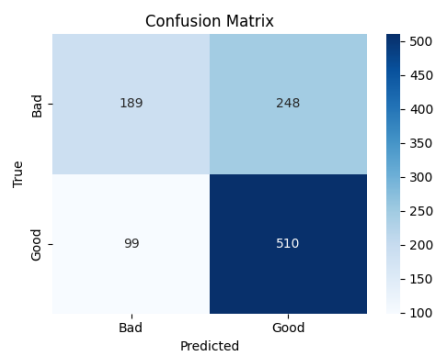


Figure A.13: Performance metrics and confusion matrix for ResNet34 Fold 1

Fold 2:

- Accuracy: 0.6631
- Precision: 0.6438
- Recall: 0.9161
- F₁-score: 0.7562
- AUC: 0.7789

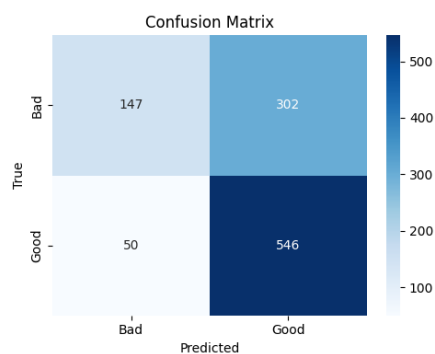


Figure A.14: Performance metrics and confusion matrix for ResNet34 Fold 2

Fold 3:

- Accuracy: 0.6956
- Precision: 0.7424
- Recall: 0.7877
- F₁-score: 0.7644
- AUC: 0.7629

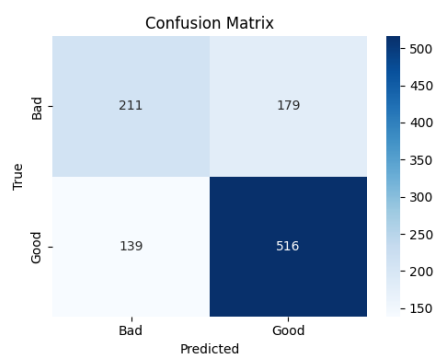


Figure A.15: Performance metrics and confusion matrix for ResNet34 Fold 3

A.2.3 DenseNet121 Results

Fold 1:

- Accuracy: 0.6615
- Precision: 0.6632
- Recall: 0.8505
- F_1 -score: 0.7453
- AUC: 0.7504

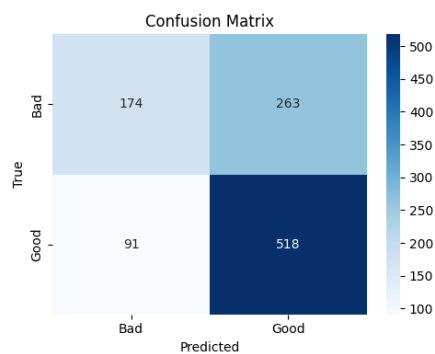


Figure A.16: Performance metrics and confusion matrix for DenseNet121 Fold

1

Fold 2:

- Accuracy: 0.6966
- Precision: 0.7136
- Recall: 0.7818
- F_1 -score: 0.7461
- AUC: 0.7586

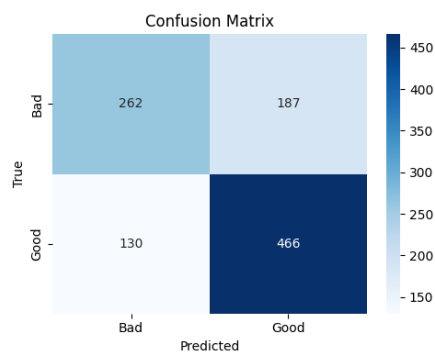


Figure A.17: Performance metrics and confusion matrix for DenseNet121 Fold

2

Fold 3:

- Accuracy: 0.7215
- Precision: 0.7577
- Recall: 0.8167
- F₁-score: 0.7861
- AUC: 0.7900

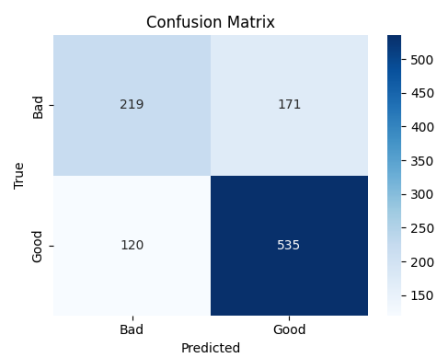


Figure A.18: Performance metrics and confusion matrix for DenseNet121 Fold

Appendix B

Temporal Classification Results

B.1 Multi-Year History Model Performance

B.1.1 Year 2 Prediction

Fold 1:

- Accuracy: 0.6534
- Precision: 0.7714
- Recall: 0.7397
- F_1 -score: 0.7552
- AUC: 0.6046

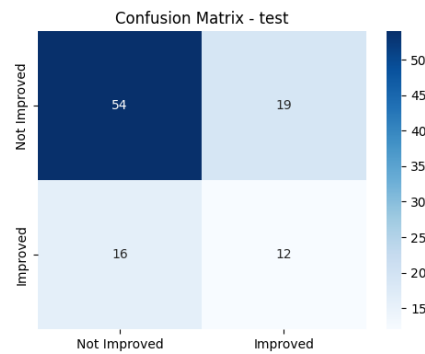


Figure B.1: Year 2 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.7227
- Precision: 0.8235
- Recall: 0.8433
- F₁-score: 0.8333
- AUC: 0.5950

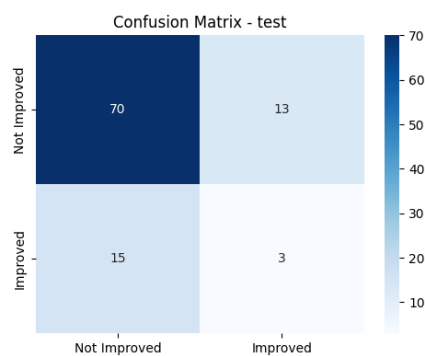


Figure B.2: Year 2 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.7200
- Precision: 0.7976
- Recall: 0.8589
- F₁-score: 0.8271
- AUC: 0.5419

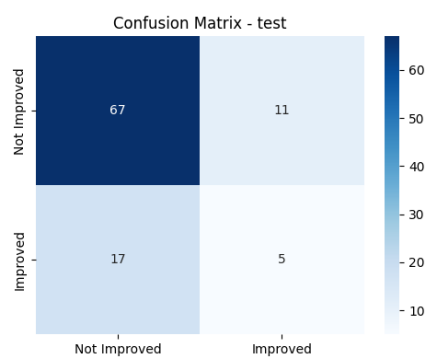


Figure B.3: Year 2 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.6534	0.7227	0.7200	0.6987 $\pm$ 0.0392
Precision	0.7714	0.8235	0.7976	0.7975 $\pm$ 0.0260
Recall	0.7397	0.8433	0.8589	0.8140 $\pm$ 0.0648
F ₁ -score	0.7552	0.8333	0.8271	0.8052 $\pm$ 0.0434
AUC	0.6046	0.5950	0.5419	0.5805 $\pm$ 0.0337

Table B.1: Year 2 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.1.2 Year 3 Prediction

Fold 1:

- Accuracy: 0.7300
- Precision: 0.7300
- Recall: 1.000
- F₁-score: 0.8439
- AUC: 0.5240

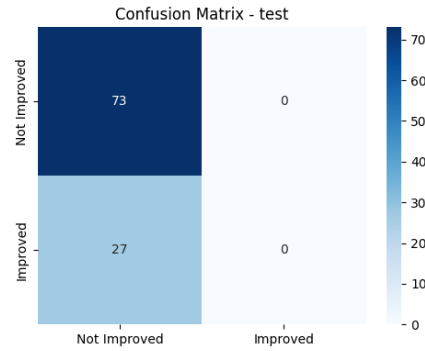


Figure B.4: Year 3 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.6600
- Precision: 0.7108
- Recall: 0.8550
- F₁-score: 0.7763
- AUC: 0.5605

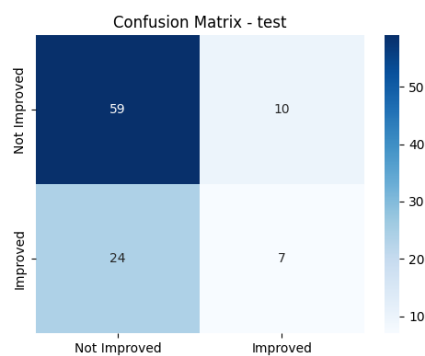


Figure B.5: Year 3 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.6363
- Precision: 0.6428
- Recall: 0.9843
- F₁-score: 0.7777
- AUC: 0.6647

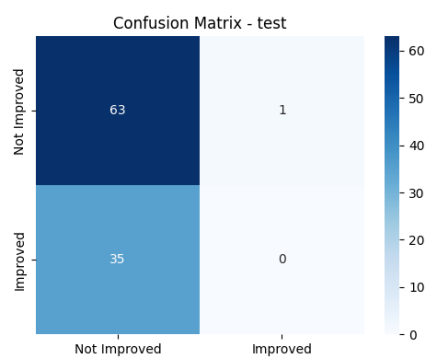


Figure B.6: Year 3 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.7300	0.6600	0.6363	0.6754 ± 0.0486
Precision	0.7300	0.7108	0.6428	0.6945 ± 0.0457
Recall	1.0000	0.8550	0.9843	0.9464 ± 0.0795
F ₁ -score	0.8439	0.7763	0.7777	0.7993 ± 0.0386
AUC	0.5240	0.5605	0.6647	0.5831 ± 0.0729

Table B.2: Year 3 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.1.3 Year 4 Prediction

Fold 1:

- Accuracy: 0.6907
- Precision: 0.6907
- Recall: 1.000
- F₁-score: 0.8170
- AUC: 0.3801

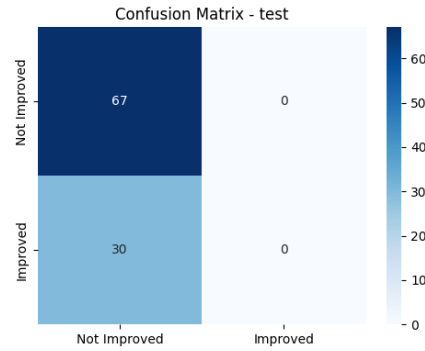


Figure B.7: Year 4 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.6875
- Precision: 0.6875
- Recall: 1.000
- F_1 -score: 0.8148
- AUC: 0.5702

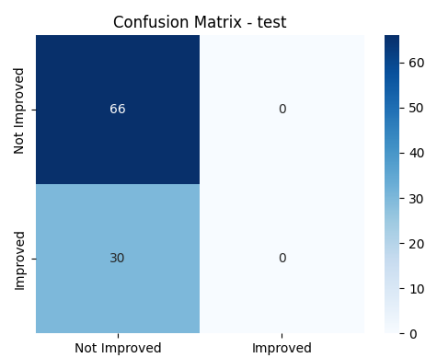


Figure B.8: Year 4 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.7604
- Precision: 0.7764
- Recall: 0.9428
- F_1 -score: 0.8516
- AUC: 0.6516

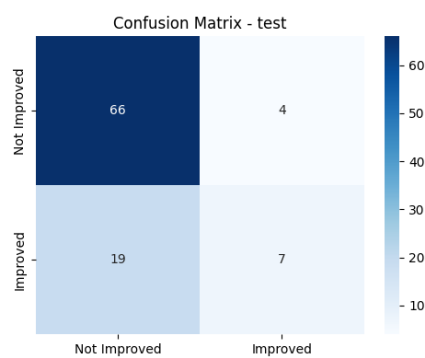


Figure B.9: Year 4 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.6907	0.6875	0.7604	0.7128 ± 0.0412
Precision	0.6907	0.6875	0.7764	0.7182 ± 0.0504
Recall	1.0000	1.000	0.9428	0.9809 ± 0.0329
F ₁ -score	0.8148	0.7763	0.8516	0.82783 ± 0.0206
AUC	0.3801	0.5702	0.6516	0.5339 ± 0.1393

Table B.3: Year 4 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.1.4 Year 5 Prediction

Fold 1:

- Accuracy: 0.7127
- Precision: 0.8000
- Recall: 0.8533
- F₁-score: 0.8258
- AUC: 0.5789

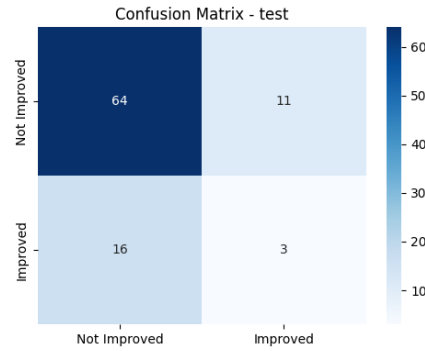


Figure B.10: Year 5 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.6276
- Precision: 0.6276
- Recall: 1.000
- F₁-score: 0.7712
- AUC: 0.4857

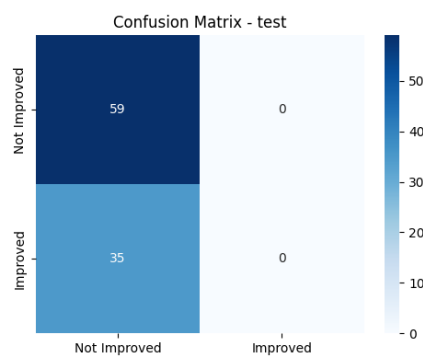


Figure B.11: Year 5 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.5957
- Precision: 0.6428
- Recall: 0.7758
- F₁-score: 0.7031
- AUC: 0.5397

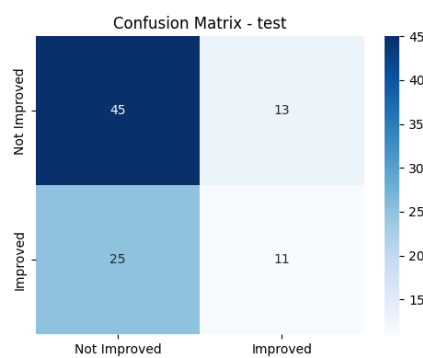


Figure B.12: Year 5 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.7127	0.6276	0.5957	0.6454 ± 0.0605
Precision	0.8000	0.6276	0.6428	0.6901 ± 0.0954
Recall	0.8533	1.000	0.7758	0.8763 ± 0.1138
F ₁ -score	0.8258	0.7712	0.7031	0.7667 ± 0.0614
AUC	0.5789	0.4857	0.5397	0.5348 ± 0.0468

Table B.4: Year 5 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.1.5 Year 6 Prediction

Fold 1:

- Accuracy: 0.6666
- Precision: 0.6630
- Recall: 1.0000
- F₁-score: 0.7973
- AUC: 0.5696

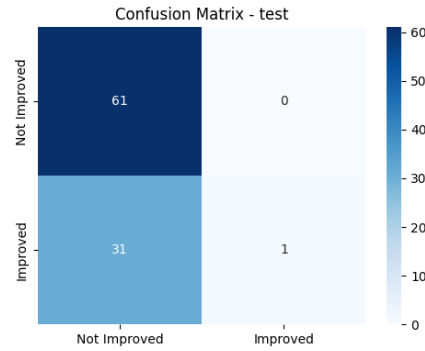


Figure B.13: Year 6 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.5434
- Precision: 0.7441
- Recall: 0.5079
- F_1 -score: 0.6037
- AUC: 0.5812

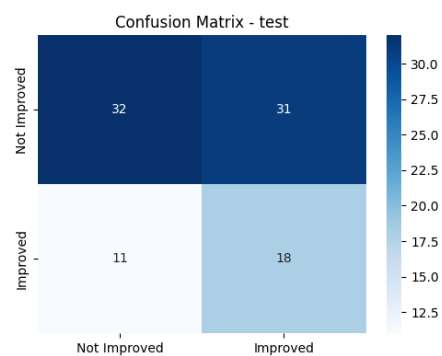


Figure B.14: Year 6 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.7173
- Precision: 0.7159
- Recall: 0.9843
- F_1 -score: 0.8289
- AUC: 0.4419

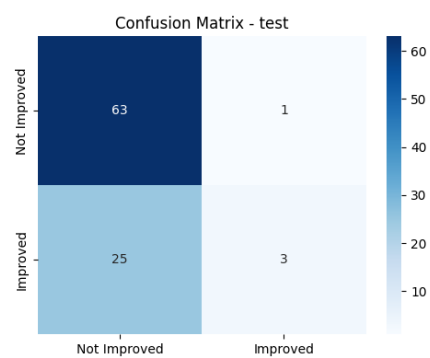


Figure B.15: Year 6 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.6666	0.5434	0.7173	0.6425 ± 0.0894
Precision	0.6630	0.7441	0.7159	0.7077 ± 0.0411
Recall	1.000	0.5079	0.9843	0.8307 ± 0.2796
F ₁ -score	0.7973	0.6037	0.8289	0.7433 ± 0.1219
AUC	0.5696	0.5812	0.4419	0.5309 ± 0.0773

Table B.5: Year 6 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.1.6 Year 7 Prediction

Fold 1:

- Accuracy: 0.7282
- Precision: 0.7441
- Recall: 0.9552
- F₁-score: 0.8366
- AUC: 0.6089

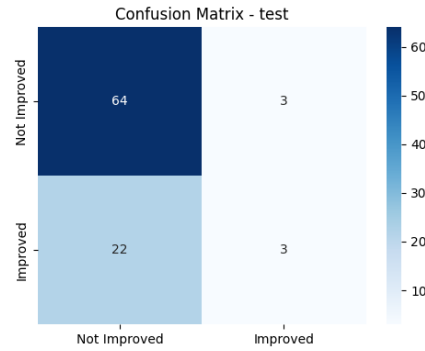


Figure B.16: Year 7 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.7500
- Precision: 0.7582
- Recall: 0.9857
- F_1 -score: 0.8571
- AUC: 0.5350

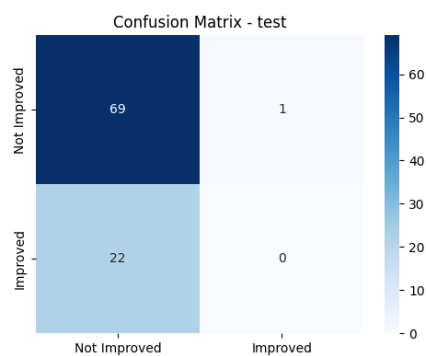


Figure B.17: Year 7 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.6923
- Precision: 0.7160
- Recall: 0.9206
- F_1 -score: 0.8055
- AUC: 0.5657

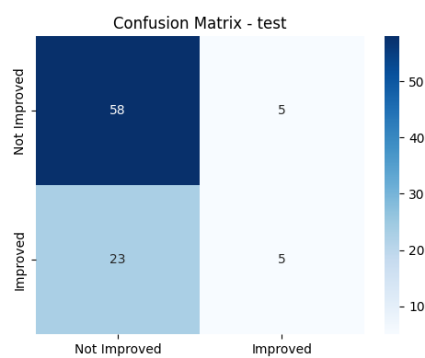


Figure B.18: Year 7 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.7282	0.7500	0.6923	0.7235 ± 0.0291
Precision	0.7441	0.7582	0.7160	0.7394 ± 0.0214
Recall	0.9552	0.9857	0.9206	0.9538 ± 0.0325
F ₁ -score	0.8366	0.8571	0.8055	0.8330 ± 0.0259
AUC	0.6089	0.5350	0.5657	0.5699 ± 0.0371

Table B.6: Year 7 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.1.7 Year 8 Prediction

Fold 1:

- Accuracy: 0.8202
- Precision: 0.8202
- Recall: 1.0000
- F₁-score: 0.9012
- AUC: 0.6558

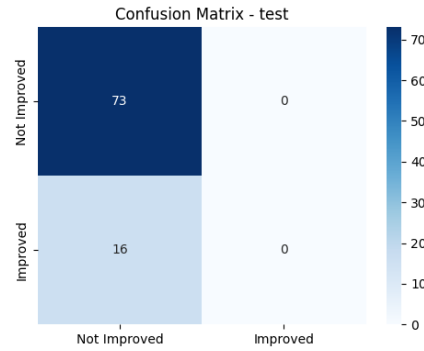


Figure B.19: Year 8 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.7272
- Precision: 0.7272
- Recall: 0.9857
- F_1 -score: 0.8421
- AUC: 0.7239

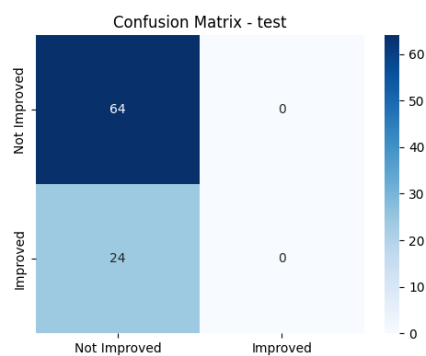


Figure B.20: Year 8 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.7386
- Precision: 0.7386
- Recall: 1.0000
- F_1 -score: 0.8496
- AUC: 0.6842

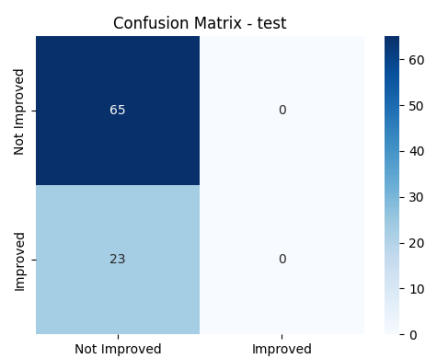


Figure B.21: Year 8 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.8202	0.7272	0.7386	0.7620 ± 0.0507
Precision	0.8202	0.7272	0.7386	0.7620 ± 0.0507
Recall	1.0000	0.9857	1.0000	1.0000 ± 0.0000
F ₁ -score	0.9012	0.8421	0.8496	0.8643 ± 0.0321
AUC	0.6558	0.7239	0.6842	0.6880 ± 0.0342

Table B.7: Year 8 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.1.8 Year 9 Prediction

Fold 1:

- Accuracy: 0.7529
- Precision: 0.7710
- Recall: 0.9696
- F₁-score: 0.8590
- AUC: 0.5446

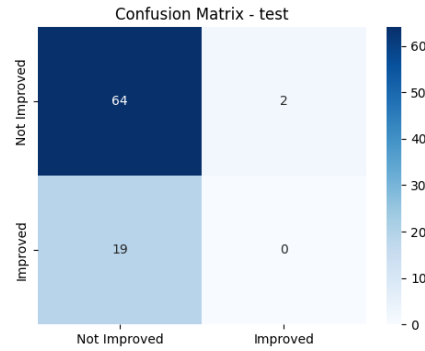


Figure B.22: Year 9 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.8470
- Precision: 0.8470
- Recall: 1.0000
- F₁-score: 0.9171
- AUC: 0.6495

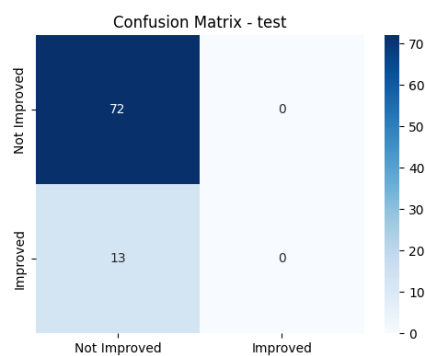


Figure B.23: Year 9 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.9761
- Precision: 0.9746
- Recall: 1.0000
- F₁-score: 0.9871
- AUC: 0.9925

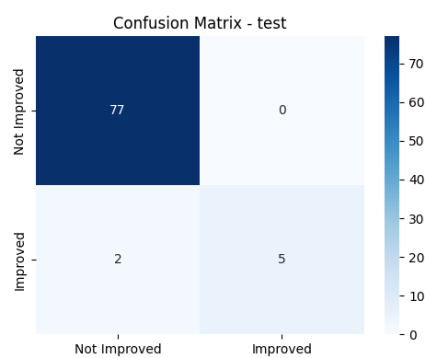


Figure B.24: Year 9 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.7529	0.8470	0.9761	0.8587 ± 0.1120
Precision	0.7710	0.8470	0.9746	0.8642 ± 0.1028
Recall	0.9696	1.0000	1.0000	0.9898 ± 0.0174
F ₁ -score	0.8590	0.9171	0.9871	0.9211 ± 0.0641
AUC	0.5446	0.6495	0.9925	0.7289 ± 0.2342

Table B.8: Year 9 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.1.9 Year 10 Prediction

Fold 1:

- Accuracy: 0.6808
- Precision: 0.7631
- Recall: 0.8285
- F₁-score: 0.7945
- AUC: 0.6761

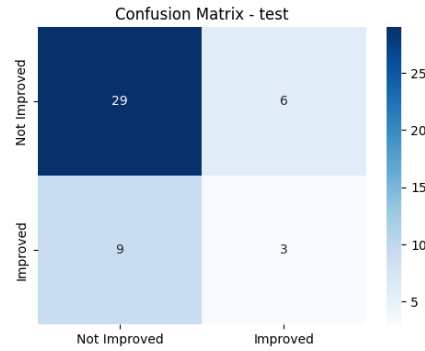


Figure B.25: Year 10 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.7021
- Precision: 0.7021
- Recall: 1.0000
- F_1 -score: 0.825
- AUC: 0.7987

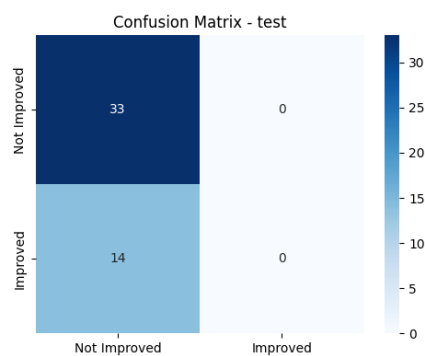


Figure B.26: Year 10 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.6304
- Precision: 0.6222
- Recall: 1.0000
- F_1 -score: 0.7671
- AUC: 0.6666

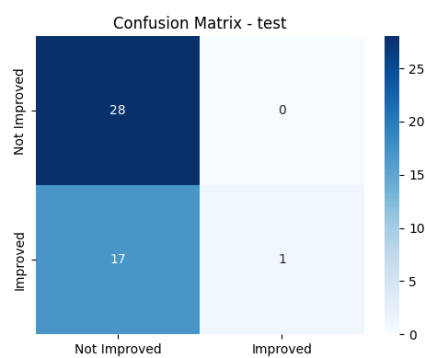


Figure B.27: Year 10 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.6808	0.7021	0.6304	0.6711 ± 0.0368
Precision	0.7631	0.7021	0.6222	0.6958 ± 0.0706
Recall	0.8285	1.0000	1.0000	0.9428 ± 0.0989
F ₁ -score	0.7945	0.8250	0.7671	0.7955 ± 0.0289
AUC	0.6761	0.7987	0.6666	0.7138 ± 0.0736

Table B.9: Year 10 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.2 Last-Year-Only Model Performance

B.2.1 Year 2 Prediction

Fold 1:

- Accuracy: 0.7227
- Precision: 0.7227
- Recall: 1.0000
- F₁-score: 0.8390
- AUC: 0.5753

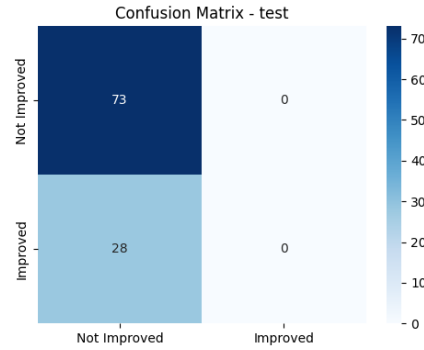


Figure B.28: Year 2 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.4950
- Precision: 0.7962
- Recall: 0.5180
- F_1 -score: 0.6277
- AUC: 0.5073

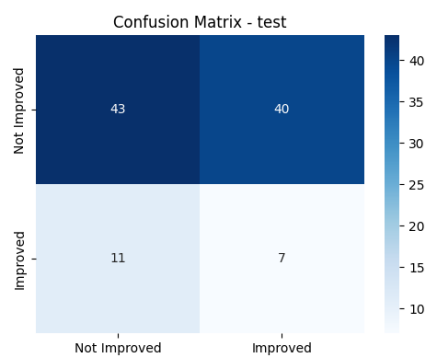


Figure B.29: Year 2 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.74
- Precision: 0.7765
- Recall: 0.9358
- F_1 -score: 0.8488
- AUC: 0.6013

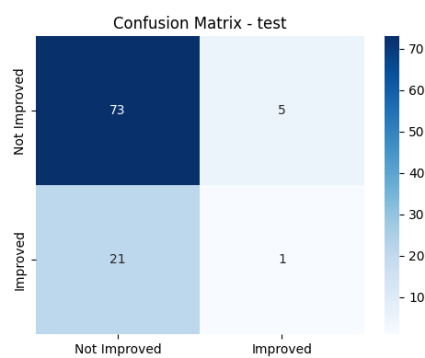


Figure B.30: Year 2 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.7227	0.4950	0.7400	0.6526 ± 0.1367
Precision	0.7227	0.7962	0.7765	0.7652 ± 0.0380
Recall	1.0000	0.5180	0.9358	0.8179 ± 0.26170
F ₁ -score	0.8390	0.6277	0.6277	0.7718 ± 0.1249
AUC	0.5753	0.5073	0.5073	0.5613 ± 0.0485

Table B.10: Year 2 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.2.2 Year 3 Prediction

Fold 1:

- Accuracy: 0.6300
- Precision: 0.7727
- Recall: 0.6986
- F₁-score: 0.7338
- AUC: 0.6007

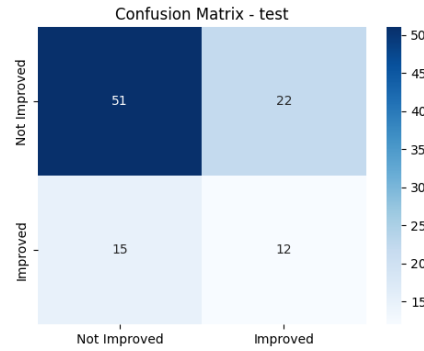


Figure B.31: Year 3 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.63
- Precision: 0.6739
- Recall: 0.8985
- F_1 -score: 0.7701
- AUC: 0.42776

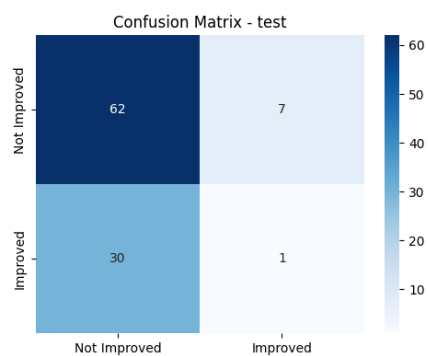


Figure B.32: Year 3 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.7070
- Precision: 0.7058
- Recall: 0.9375
- F_1 -score: 0.8053
- AUC: 0.6946

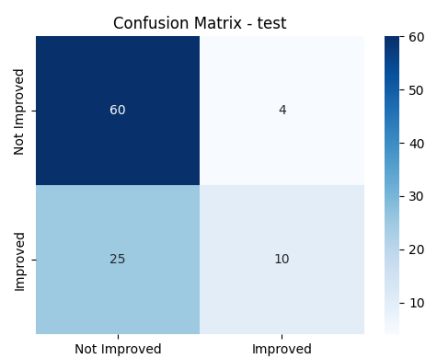


Figure B.33: Year 3 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.6300	0.6300	0.7070	0.6556 ± 0.0444
Precision	0.7727	0.6739	0.7058	0.7175 ± 0.0504
Recall	0.6986	0.8985	0.9375	0.8448 ± 0.1281
F ₁ -score	0.7338	0.7701	0.8053	0.7697 ± 0.0357
AUC	0.6007	0.4277	0.6946	0.5743 ± 0.1353

Table B.11: Year 3 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.2.3 Year 4 Prediction

Fold 1:

- Accuracy: 0.6800
- Precision: 0.6800
- Recall: 1.000
- F₁-score: 0.8095
- AUC: 0.4641

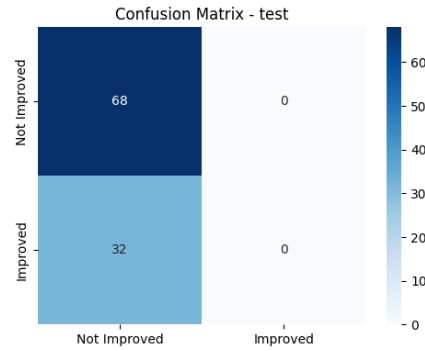


Figure B.34: Year 4 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.7700
- Precision: 0.8923
- Recall: 0.7837
- F₁-score: 0.8345
- AUC: 0.8487

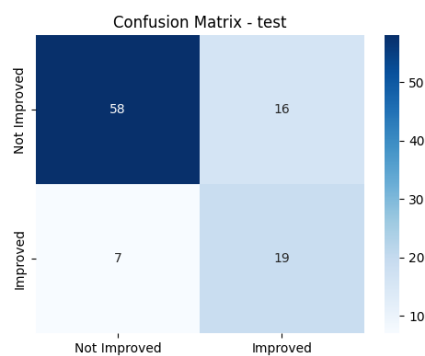


Figure B.35: Year 4 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.6262
- Precision: 0.6630
- Recall: 0.9104
- F₁-score: 0.7672
- AUC: 0.4365

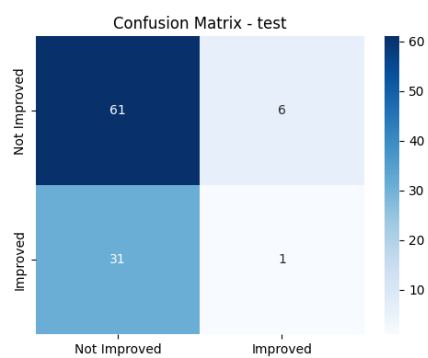


Figure B.36: Year 4 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.6800	0.7700	0.6262	0.6920 ± 0.0726
Precision	0.6800	0.8923	0.6630	0.7451 ± 0.1277
Recall	1.0000	0.7837	0.9104	0.8980 ± 0.1086
F ₁ -score	0.8095	0.8345	0.7672	0.8037 ± 0.0339
AUC	0.4641	0.8487	0.4365	0.5831 ± 0.2304

Table B.12: Year 4 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.2.4 Year 5 Prediction

Fold 1:

- Accuracy: 0.7676
- Precision: 0.7676
- Recall: 1.0000
- F₁-score: 0.8685
- AUC: 0.6624

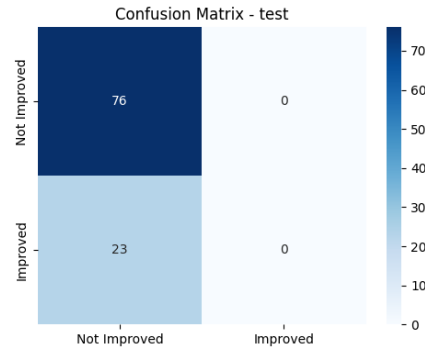


Figure B.37: Year 5 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.6836
- Precision: 0.6836
- Recall: 1.000
- F₁-score: 0.8121
- AUC: 0.5089

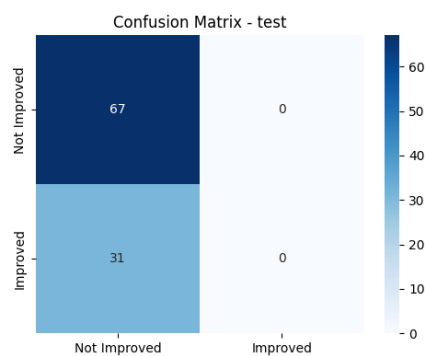


Figure B.38: Year 5 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.6224
- Precision: 0.6224
- Recall: 1.000
- F₁-score: 0.7672
- AUC: 0.5866

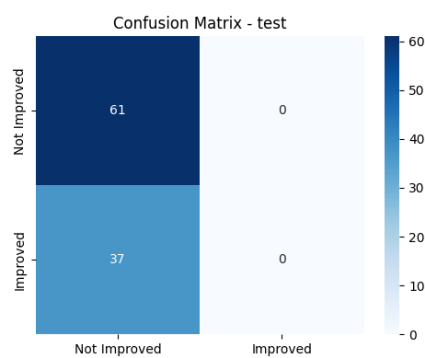


Figure B.39: Year 5 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.7676	0.6836	0.6224	0.6912 ± 0.0729
Precision	0.7676	0.6836	0.6224	0.6912 ± 0.0729
Recall	1.0000	1.000	1.0000	1.0000 ± 0.0000
F ₁ -score	0.8685	0.8121	0.7672	0.8159 ± 0.0507
AUC	0.6624	0.5089	0.5866	0.5859 ± 0.0767

Table B.13: Year 5 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.2.5 Year 6 Prediction

Fold 1:

- Accuracy: 0.5900
- Precision: 0.7187
- Recall: 0.6666
- F₁-score: 0.6917
- AUC: 0.5792

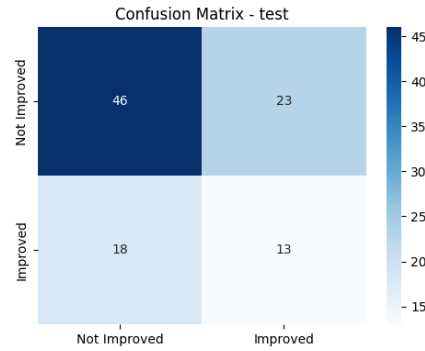


Figure B.40: Year 6 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.7300
- Precision: 0.7300
- Recall: 1.0000
- F₁-score: 0.8439
- AUC: 0.5459

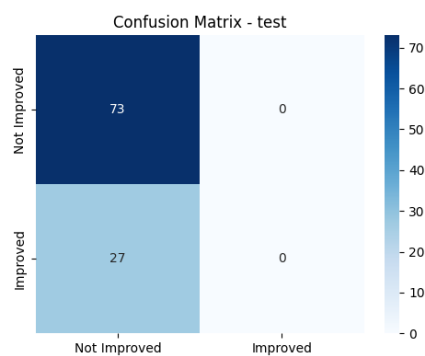


Figure B.41: Year 6 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.6363
- Precision: 0.6363
- Recall: 1.0000
- F₁-score: 0.7777
- AUC: 0.8505

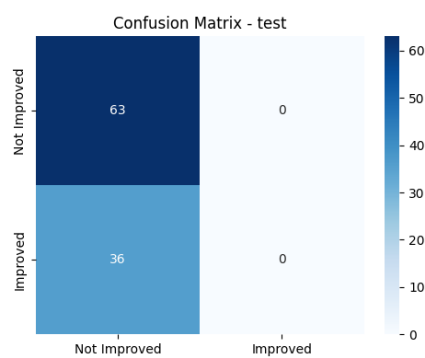


Figure B.42: Year 6 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.5900	0.7300	0.6363	0.6521 ± 0.0713
Precision	0.7187	0.7300	0.6363	0.6950 ± 0.0511
Recall	0.6666	1.0000	1.0000	0.8888 ± 0.1924
F ₁ -score	0.6917	0.8439	0.7777	0.7711 ± 0.0763
AUC	0.5792	0.5459	0.5459	0.6585 ± 0.1670

Table B.14: Year 6 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.2.6 Year 7 Prediction

Fold 1:

- Accuracy: 0.6601
- Precision: 0.6888
- Recall: 0.8985
- F₁-score: 0.7798
- AUC: 0.5532

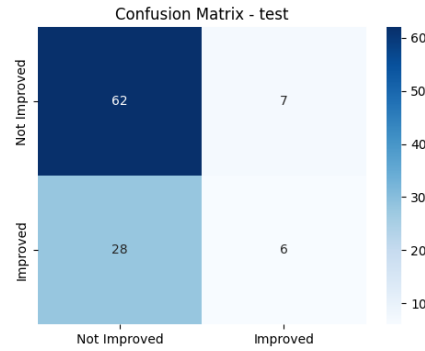


Figure B.43: Year 7 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.8627
- Precision: 0.8541
- Recall: 1.0000
- F₁-score: 0.9213
- AUC: 0.9609

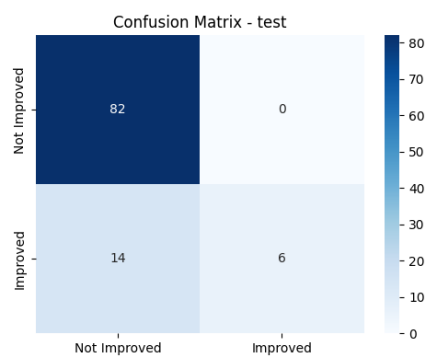


Figure B.44: Year 7 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.7352
- Precision: 0.7474
- Recall: 0.9736
- F₁-score: 0.8457
- AUC: 0.6968

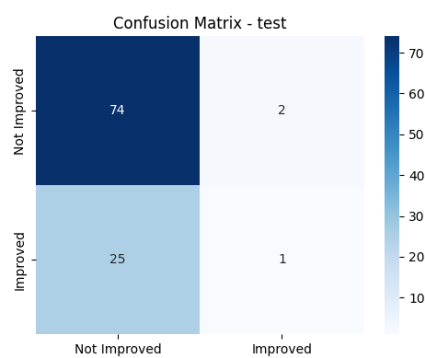


Figure B.45: Year 7 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.6601	0.8627	0.7352	0.7527 ± 0.1023
Precision	0.6888	0.8541	0.7474	0.7635 ± 0.0837
Recall	0.8985	1.0000	0.9736	0.9574 ± 0.0526
F ₁ -score	0.7798	0.9213	0.8457	0.8489 ± 0.0707
AUC	0.5532	0.9609	0.6968	0.7370 ± 0.2067

Table B.15: Year 7 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.2.7 Year 8 Prediction

Fold 1:

- Accuracy: 0.7700
- Precision: 0.7700
- Recall: 1.0000
- F₁-score: 0.8700
- AUC: 0.6860

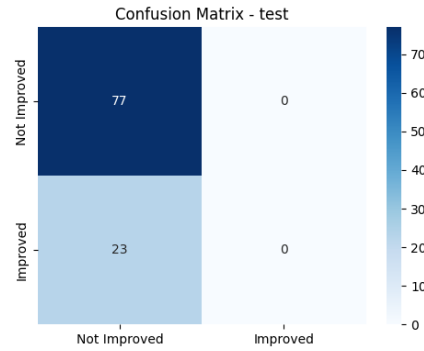


Figure B.46: Year 8 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.8100
- Precision: 0.8100
- Recall: 1.0000
- F_1 -score: 0.8950
- AUC: 0.8037

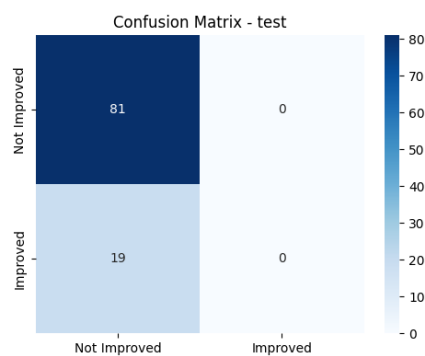


Figure B.47: Year 8 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.8500
- Precision: 0.8351
- Recall: 1.0000
- F_1 -score: 0.9101
- AUC: 0.8919

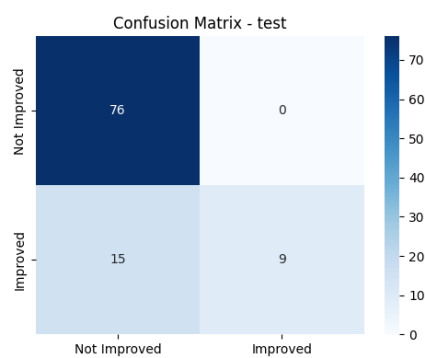


Figure B.48: Year 8 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.7700	0.8100	0.8500	0.8100 ± 0.0400
Precision	0.7700	0.8100	0.8351	0.8050 ± 0.0328
Recall	1.0000	1.0000	1.0000	1.0000 ± 0.0000
F ₁ -score	0.8700	0.8950	0.9101	0.8917 ± 0.0202
AUC	0.6860	0.8037	0.8919	0.7939 ± 0.1033

Table B.16: Year 8 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.2.8 Year 9 Prediction

Fold 1:

- Accuracy: 0.8333
- Precision: 0.8333
- Recall: 1.0000
- F₁-score: 0.9090
- AUC: 0.5781

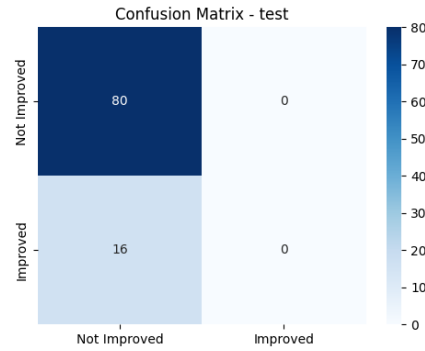


Figure B.49: Year 9 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.8421
- Precision: 0.8421
- Recall: 1.0000
- F_1 -score: 0.9142
- AUC: 0.4183

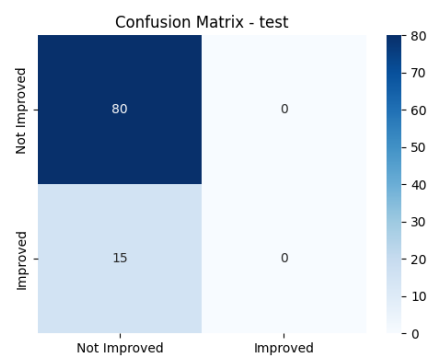


Figure B.50: Year 9 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.8315
- Precision: 0.9024
- Recall: 0.9024
- F_1 -score: 0.9024
- AUC: 0.7636

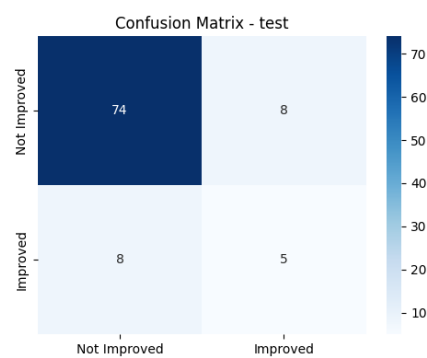


Figure B.51: Year 9 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.8333	0.8421	0.8315	0.8356 ± 0.0056
Precision	0.8333	0.8421	0.9024	0.8592 ± 0.0376
Recall	1.0000	1.0000	0.9024	0.9674 ± 0.0563
F ₁ -score	0.9090	0.9142	0.9024	0.9086 ± 0.00593
AUC	0.5781	0.4183	0.7636	0.5866 ± 0.1727

Table B.17: Year 9 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

B.2.9 Year 10 Prediction

Fold 1:

- Accuracy: 0.7962
- Precision: 0.7872
- Recall: 0.9736
- F₁-score: 0.8705
- AUC: 0.7697

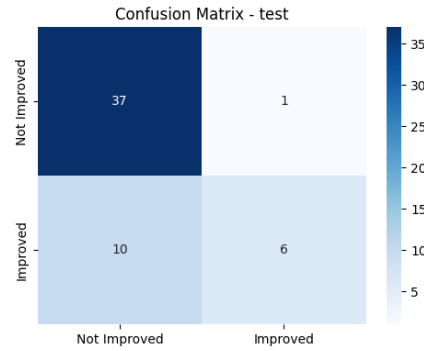


Figure B.52: Year 10 - Performance metrics and confusion matrix for Fold 1

Fold 1:

- Accuracy: 0.7037
- Precision: 0.7200
- Recall: 0.9473
- F_1 -score: 0.8181
- AUC: 0.7351

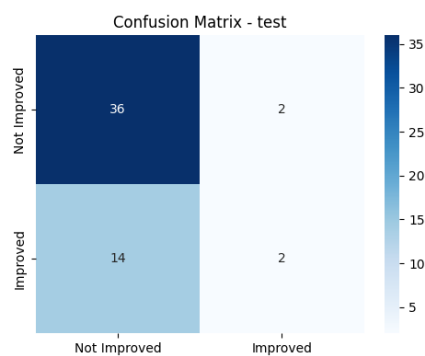


Figure B.53: Year 10 - Performance metrics and confusion matrix for Fold 2

Fold 1:

- Accuracy: 0.6666
- Precision: 0.6666
- Recall: 1.0000
- F_1 -score: 0.8000
- AUC: 0.6404

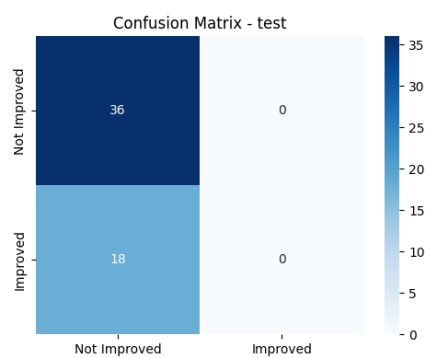


Figure B.54: Year 10 - Performance metrics and confusion matrix for Fold 3

Metrics	Fold 1	Fold 2	Fold 3	Mean $\pm$ std
Accuracy	0.7962	0.7037	0.6666	0.7222 ± 0.0667
Precision	0.7872	0.7200	0.6666	0.7246 ± 0.0604
Recall	0.9736	0.9473	1.0000	0.9736 ± 0.0263
F ₁ -score	0.8705	0.8181	0.8000	0.8295 ± 0.0366
AUC	0.7697	0.7351	0.6404	0.7151 ± 0.0669

Table B.18: Year 10 - Performance across 3-fold cross-validation (mean $\pm$ standard deviation)

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